

*A Minor Project Report
on*

CANTEEN PRE-ORDERING SYSTEM FOR CAMPUS FOOD COURTS

submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

in

CSE (Data Science)

by

N Venkata Aashrith (24211A6776)

Under the guidance of

MS S. Durgadevi, M. Tech

Assistant Professor



DEPARTMENT OF CSE (Data Science)

B. V. RAJU INSTITUTE OF TECHNOLOGY

(UGC Autonomous, Accredited by NBA & NAAC)

Vishnupur, Narsapur, Medak (Dist.), Telangana State, India - 502313

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B. V. Raju Institute of Technology

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Vishnupur, Narsapur, Medak (Dist.),

Telangana State, India – 502313

DEPARTMENT OF CSE (Data Science)

CERTIFICATE

This is to certify that the Minor Project entitled “**Canteen Pre-Ordering System for Campus Food Courts**”, being submitted by

N Venkata Aashrith (24211A6776)

In partial fulfillment of the requirements for the award of degree of BACHELOR OF TECHNOLOGY in **CSE (Data Science)** to B.V. RAJU INSTITUTE OF TECHNOLOGY is a record of Bonafide work carried out during a period from **December 2025 to May 2026** by them under the guidance of **MS S. Durgadevi**, Assistant Professor, **CSE (Data Science)** Department.

This is to certify that the above statement made by the students is/are correct to the best of my knowledge.

MS S. Durgadevi
Assistant Professor

The **Minor Project** Viva-Voce Examination of this team has been held on _____.

External Examiner

HOD-CSE (Data Science)



B. V. Raju Institute of Technology

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Vishnupur, Narsapur, Medak (Dist.),

Telangana State, India – 502313

CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the project entitled **“Canteen Pre-Ordering System for Campus Food Courts”** in partial fulfillment of the requirements for the award of Degree of Bachelor of Technology and submitted in the **Department of CSE (Data Science)**, B. V. Raju Institute of Technology, Narsapur is an authentic record of my own work carried out during a period from **December 2025 to May 2026** under the guidance of **MS S Durgadevi**, Assistant Professor. The work presented in this project report has not been submitted by me for the award of any other degree of this or any other Institute/University.

N Venkata Aashrith (24211A6776)

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N Venkata Aashrith (24211A6776)

CANTEEN PRE-ORDERING SYSTEM FOR CAMPUS FOOD COURTS

ABSTRACT

The Canteen Pre-Ordering System for Campus Food Courts is an Android-based mobile application developed using Android Studio with Kotlin programming language. The system is designed to streamline the food ordering process at B.V. Raju Institute of Technology (BVRIT), specifically targeting the two campus canteens — Daffodils 2 (D2) and Food Court (FC). The primary objective of this application is to eliminate the problem of long queues, reduce waiting time, and improve the overall dining experience for students and faculty members.

The application allows registered users to browse the available menu items from either canteen, select their desired food items, place pre-orders, and receive their food at the designated time without waiting in queues. The system incorporates user authentication, a dynamic menu display, cart management, order placement, and order tracking functionalities. The backend operations are supported by Firebase Realtime Database and Firebase Authentication, which ensure secure, real-time data management.

The app has been partially deployed and tested within the campus environment. The canteen staff can view incoming orders through an admin panel and prepare food accordingly, thereby optimizing kitchen operations. The solution addresses the inefficiencies in the traditional paper-based or walk-in food ordering system and replaces it with a fast, convenient, and scalable digital solution.

The system is developed using modern Android development practices with a single-activity, state-driven architecture implemented through Jetpack Compose. A modular design approach

is followed to improve maintainability and scalability. The documentation covers all major phases of the Software Development Life Cycle (SDLC), including requirements analysis, system design, implementation, and testing.

Keywords: Android Application, Kotlin, Canteen Pre-Ordering, Food Court, Firebase, Campus Food Management, Mobile Application, MVVM Architecture, BVRIT

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CHAPTER 1

INTRODUCTION

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INTRODUCTION

1.1 Motivation

In today's fast-paced academic environment, efficiency and convenience have become essential factors in enhancing student life on campus. At B.V. Raju Institute of Technology (BVRIT), canteen facilities such as Daffodils 2 (D2) and Food Court (FC) serve a large population of students and faculty daily. During peak hours, especially lunch breaks, these canteens experience heavy crowding, resulting in long queues and increased waiting times. Students often spend a significant portion of their limited break time standing in line rather than utilizing it for meals, rest, or academic activities.

The rapid growth of smartphone usage among students presents an opportunity to modernize traditional systems. Mobile applications have transformed many industries by enabling users to access services quickly and efficiently. Applying this concept to campus canteens can significantly improve both user experience and operational efficiency.

A mobile-based canteen pre-ordering system allows users to browse available menu items, place orders in advance, and collect food at the appropriate time. This eliminates the need for physical queuing and reduces congestion at the counter. Additionally, such a system provides real-time updates regarding order status, ensuring transparency and reducing uncertainty for users.

From the perspective of canteen staff, a digital system offers the advantage of receiving orders in advance, allowing better planning and preparation. This leads to improved resource management, reduced food wastage, and smoother workflow during peak hours.

Furthermore, this project serves as a valuable learning platform for implementing real-world technologies. It involves Android application development using Kotlin, modern UI design with Jetpack Compose, and backend integration using Firebase services such as Authentication and Firestore. These technologies are widely used in the industry, making the project relevant from both academic and professional perspectives.

Thus, the motivation behind this project is to create a smart, efficient, and user-friendly system that improves the overall canteen experience while leveraging modern mobile and cloud technologies.

1.2 Problem Definition

The current food ordering system at BVRIT canteens operates entirely on a manual basis. While simple, this system suffers from multiple inefficiencies that negatively affect both users and canteen staff. The key problems identified are as follows:

1.2.1 Long Waiting Time

During peak hours, students are required to stand in long queues to place their orders. The waiting time can extend to 20–30 minutes, which significantly reduces the time available for eating and relaxation within a short break period.

1.2.2 Uncertainty in Food Availability

Students have no prior information about the availability of menu items. As a result, they may wait in line only to discover that their preferred food item is unavailable, leading to frustration and wasted time.

1.2.3 Lack of Pre-Planning for Staff

Since orders are received only at the counter, canteen staff have no advance knowledge of demand. This makes it difficult to plan food preparation effectively, often resulting in either shortages or excess production.

1.2.4 Lack of Order Transparency

After placing an order, users receive no updates regarding its preparation status. They are required to wait near the counter, leading to overcrowding and confusion.

1.2.5 Absence of Digital Records

The existing system does not maintain any digital records of orders. This prevents users from accessing their order history and restricts management from analyzing trends such as popular items and peak hours.

1.2.6 Manual Payment System

All transactions are handled manually using cash. This increases transaction time, creates dependency on exact change, and may lead to calculation errors.

Problem Statement

The absence of a digital pre-ordering system in campus canteens leads to long waiting times, inefficient order handling, lack of transparency, and poor user experience.

1.3 Objective of Project

The primary objective of this project is to develop a **Smart Canteen Android Application** that provides an efficient and user-friendly platform for food ordering and management.

1.3.1 Develop a User-Friendly Android Application

Design and implement a mobile application using Kotlin and Jetpack Compose that offers an intuitive interface and smooth navigation for users.

1.3.2 Enable Multi-Canteen Support (D2 & FC)

Provide support for both Daffodils 2 (D2) and Food Court (FC), allowing users to select their preferred canteen and access its menu.

1.3.3 Implement Secure Authentication

Use Firebase Phone Authentication with OTP verification to ensure secure and reliable user login.

1.3.4 Real-Time Menu Management

Integrate Firebase Firestore to display menu items dynamically, ensuring that users always see updated information regarding availability and pricing.

1.3.5 Cart and Order Management

Allow users to add items to the cart, adjust quantities, include notes, and place orders. Each order should generate a unique identifier and be stored securely.

1.3.6 Staff Order Management System

Provide functionality for staff to view incoming orders and update their status (Placed, Preparing, Ready) in real time.

1.3.7 Real-Time Order Tracking

Enable users to track their orders through real-time updates using Firestore listeners.

1.3.8 Maintain Order History

Store all user orders in the database, allowing users to view past orders and provide feedback.

1.3.9 Reduce Waiting Time and Queue Length

Minimize physical crowding by allowing users to place orders in advance.

1.3.10 Ensure Scalability and Maintainability

Design the system using modular architecture principles to allow easy future enhancements and expansion.

1.4 Limitations of Project

While the Canteen Pre-Ordering System addresses several key problems in campus food management, the following limitations have been identified in the current version of the application:

- The application is currently designed exclusively for Android devices. iOS support has not been developed in this version due to the additional requirement of macOS and Xcode development environment.
- Online payment gateway integration (UPI, net banking, debit/credit card) has not been implemented in this version. The current system supports order placement with cash payment at the time of pickup. Digital payment can be added in future iterations.
- The system requires a stable internet connection to function, as it relies entirely on Firebase Realtime Database for data storage and retrieval. Offline functionality is not supported.
- The application has been partially deployed and tested within the campus environment. Full-scale deployment covering all menu items of both canteens (D2 and FC) is yet to be completed.
- The application currently does not support push notifications for order status updates. Users must manually refresh or navigate to the order tracking screen to check their order status.
- The system does not currently integrate with the college's student information system or ID card database. User verification is limited to email-based registration without institutional email domain enforcement.
- The application does not support table reservation or seating management functionality. It is solely focused on food pre-ordering and pickup.
- Real-time inventory management that automatically adjusts available quantities as orders are placed has not been fully implemented. Canteen admins must manually update item availability.
- The application has been tested on a limited number of Android devices and may not be fully optimized for all screen sizes and resolutions.
- The current admin panel is a basic interface suitable for limited canteen operations. Advanced analytics, revenue dashboards, and inventory reports are not included in this version.

1.5 Organization of Documentation

This documentation is organized into nine chapters, each covering a distinct phase or aspect of the Canteen Pre-Ordering System project. The structure is as follows:

- Chapter 1 - Introduction: Provides the project background including motivation, problem definition, objectives, limitations, and the organization of this documentation.

- Chapter 2 - Literature Survey: Reviews existing food ordering systems and related work, analyzes the disadvantages of existing systems, and presents the proposed system as the solution.
- Chapter 3 - Analysis: Covers the system requirements analysis including functional and non-functional requirements, software and hardware requirements, architecture diagrams, algorithms, and flowcharts.
- Chapter 4 - Design: Details the system design including module descriptions, Data Flow Diagrams (DFD), Entity-Relationship Diagram (ERD), UML diagrams, and the module design and organization.
- Chapter 5 - Output Screenshots and Description: Presents the actual screens of the developed Android application along with descriptions of each screen's functionality.
- Chapter 6 - Sample Code: Contains representative code snippets from the Kotlin codebase covering key modules such as authentication, menu loading, cart management, and order placement.
- Chapter 7 - Conclusion: Summarizes the project outcomes, what was achieved, and the impact of the system.
- Chapter 8 - Future Work: Outlines potential enhancements and features that can be added to the system in future development cycles.
- Chapter 9 - References and Citations: Lists all academic papers, books, online resources, and documentation referenced during the development of this project.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Digital food ordering systems have become an essential part of modern service environments due to the growing demand for convenience, speed, and efficiency. With the increasing penetration of smartphones and internet connectivity, mobile-based applications have transformed the way users interact with food service systems. Institutions such as universities, corporate campuses, and hospitals have started adopting digital solutions to streamline their food ordering processes and improve user satisfaction.

Several research studies highlight that implementing a pre-ordering system significantly reduces waiting time and enhances service efficiency. In traditional systems, users are required to stand in queues to place and receive orders, which leads to time wastage, especially during peak hours. Digital systems eliminate this issue by allowing users to place orders remotely and collect them when ready.

Commercial platforms like Swiggy and Zomato have proven the success of mobile-based food ordering. These systems provide features such as menu browsing, real-time order tracking, and digital payments. However, they are designed for large-scale restaurant ecosystems and are not suitable for closed environments like college campuses, where user base, menu structure, and operational requirements are different.

In recent years, Firebase has emerged as a popular backend solution for academic and small-scale applications. It provides real-time database capabilities, authentication services, and cloud storage, making it ideal for applications requiring instant data synchronization. Similarly, Kotlin has become the preferred programming language for Android development due to its concise syntax, improved safety features, and support for modern development practices such as asynchronous programming.

This project leverages these technologies to develop a Smart Canteen System tailored specifically for BVRIT canteens — Daffodils 2 (D2) and Food Court (FC). The system focuses on providing real-time updates, efficient order management, and an improved user experience.

2.2 Existing System

The current food ordering system at BVRIT canteens operates using a completely manual approach. While simple, this system lacks efficiency and scalability, especially during peak hours.

2.2.1 Walk-In Ordering

In the existing system, students physically visit the canteen and stand in queues to place their orders. The order is communicated verbally to the staff, after which the student waits for preparation and collection. This process is time-consuming and inefficient, particularly when the number of customers is high.

2.2.2 Menu Display System

Menu items are displayed using physical boards or charts within the canteen premises. These boards are updated manually and may not always reflect the current availability of items. As a result, students often face situations where items listed on the board are actually unavailable.

2.2.3 Cash-Based Payments

All transactions are handled manually using cash. This not only slows down the ordering process but also increases the chances of errors in billing and handling change. During busy hours, payment processing becomes a major bottleneck.

2.2.4 No Order Tracking

Once an order is placed, the student receives no updates regarding its status. They are required to wait near the counter until their order is ready, leading to crowding and confusion.

Comparison of Existing Systems

System	Pre-Order	Campus Specific	Real-Time Status	Mobile App	Freeto Deploy
Swiggy/Zomato	Yes	No	Yes	Yes	No
Manual Counter	No	Yes	No	No	Yes
Google Forms	Partial	Yes	No	No	Yes
Proposed System	Yes	Yes	Yes	Yes	Yes

2.3 Disadvantages of Existing System

The manual system currently used in the canteens has several drawbacks:

- **Time In efficiency:**

Students spend a significant amount of time waiting in queues, especially during peak hours, reducing their effective break time.

- **No Menu Preview:**

There is no option to check available items before visiting the canteen, leading to inconvenience if items are unavailable.

- **Order Errors:**

Since orders are communicated verbally, there is a high chance of misunderstandings between customers and staff.

- **No Data Storage:**

The system does not maintain any digital records of orders, making it impossible to analyze trends or maintain history.

- **Scalability Issues:**

As the number of users increases, the system becomes more inefficient and difficult to manage.

- **Lack of Personalization:**

There are no user-specific features such as saved preferences or order history.

- **Operational Challenges:**

Staff cannot predict demand in advance, leading to either over-preparation or shortage of food.

- **Paper Usage:**

The use of paper tokens contributes to unnecessary waste and environmental impact.

2.4 Proposed System

To overcome the limitations of the existing system, a Smart Canteen Android Application is proposed.

2.4.1 Digital Pre-Ordering

The system allows users to browse menu items and place orders in advance using their smartphones. This eliminates the need to stand in queues and reduces waiting time significantly.

2.4.2 Real-Time Order Tracking

Users can track their orders in real time through the application. The order status is updated dynamically (Placed, Preparing, Ready), allowing users to arrive at the canteen only when their order is ready.

2.4.3 Dual Canteen Support (D2 & FC)

The application supports both Daffodils 2 (D2) and Food Court (FC), providing separate menus and order management for each canteen. This ensures accurate handling of orders based on location.

2.4.4 Firebase Integration

The system uses Firebase services for backend operations:

- Authentication: OTP-based login
- Firestore: Real-time database for orders and menu
- Storage: Image handling

This ensures scalability, reliability, and real-time data synchronization.

2.4.5 Staff and Admin Panel

A dedicated interface is provided for staff to manage orders and update their status. Admins can add, edit, or remove menu items and control availability. This improves operational efficiency and reduces manual workload.

2.4.6 Order History and Profile Management

Users can view their past orders and manage their profile information within the app. This enhances user experience by enabling quick reordering and personalization.

CHAPTER 3

ANALYSIS

CHAPTER – 3

ANALYSIS

3.1 Introduction

System analysis is an essential phase in the software development lifecycle, focusing on understanding the problem, identifying user needs, and defining system requirements. For the Smart Canteen Pre-Ordering System, analysis was carried out through observation of canteen operations, interaction with students, and evaluation of current ordering methods.

The system is designed to serve two primary user groups:

- Students: Users who browse menus, place orders, and track order status
- Canteen Staff/Admin: Users who manage menu items and update order status

An additional stakeholder is the institution (BVRIT), which benefits from improved service efficiency and reduced congestion in canteens.

The student interaction flow includes:

- OTP-based login
- Location selection (D2 / FC)
- Menu browsing
- Cart management
- Order placement
- Order tracking

The staff interaction flow includes:

- Login
- Viewing active orders
- Updating order status (Placed → Preparing → Ready)
- Managing menu availability

The technology stack was selected based on simplicity, scalability, and real-time capabilities:

- Android (Kotlin + Jetpack Compose) for frontend
- Firebase (Authentication + Firestore + Storage) for backend

This ensures a serverless architecture with minimal maintenance and real-time synchronization.

3.2 Software Requirement Specification (SRS)

The SRS defines the functional and non-functional requirements of the system.

3.2.1 User Requirements

Student Requirements

1. The system shall allow users to log in using phone number and OTP.
2. The system shall allow users to select canteen location (D2 or FC).
3. The system shall display menu items based on selected location.
4. The system shall allow users to search and filter items.
5. The system shall allow adding items to cart with quantity and notes.
6. The system shall allow users to place orders.
7. The system shall show real-time order status updates.
8. The system shall maintain order history.
9. The system shall allow users to submit feedback and ratings.

Staff/Admin Requirements

10. The system shall allow staff/admin login.
11. The system shall allow adding, updating, and deleting menu items.
12. The system shall display active orders in real time.
13. The system shall allow updating order status.
14. The system shall allow toggling item availability.
15. The system shall allow viewing customer feedback.

Functional Requirements Table

FR ID	Requirement Description	User Role	Priority
FR-01	Phone OTP authentication	Student	High

FR ID	Requirement Description	User Role	Priority
FR-02	Location selection (D2/FC)	Student	High
FR-03	Browse menu items	Student	High
FR-04	Add to cart	Student	High
FR-05	Place order	Student	High
FR-06	Real-time order tracking	Both	High
FR-07	Order history	Student	Medium
FR-08	Menu management	Admin	High
FR-09	Order status update	Staff	High
FR-10	Feedback system	Student	Low

Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	App should load within 3 seconds
NFR-02	Performance	Order placement < 5 seconds
NFR-03	Security	OTP-based authentication
NFR-04	Usability	Easy to use without training
NFR-05	Availability	System available during canteen hours
NFR-06	Scalability	Support multiple users simultaneously
NFR-07	Compatibility	Android 8.0 and above

NFR ID	Category	Requirement
NFR-08	Maintainability	Modular architecture
NFR-09	Reliability	Handle network errors gracefully

3.2.2 Software Requirements

Component	Technology
Language	Kotlin
IDE	Android Studio
Backend	Firebase
Database	Firestore
Authentication	Firebase OTP
Storage	Firebase Storage
UI	Jetpack Compose
Architecture	MVVM
Version Control	Git

3.2.3 Hardware Requirements

Category	Minimum	Recommended
System	i5, 8GB RAM	i7, 16GB RAM
OS	Windows 10	Windows 11

Category	Minimum	Recommended
Device	Android 8.0	Android 11+
Internet	3G	4G/WiFi

3.3 Architecture Diagram and Description

The System Architecture Diagram of the BVRIT Smart Canteen System represents the overall structure of the application, showing how different layers interact with each other to provide a seamless food ordering experience.

The system follows a three-layer architecture, consisting of the Client Layer, Application Logic Layer, and Backend Layer.

The *Client Layer* includes three types of Android applications: Student App, Staff App, and Admin App. Students can browse menu items, place orders, and track their orders. Staff members can view incoming orders and update their status, while administrators can manage users, menu items, and generate reports. Navigation between screens is handled using a state-based navigation system.

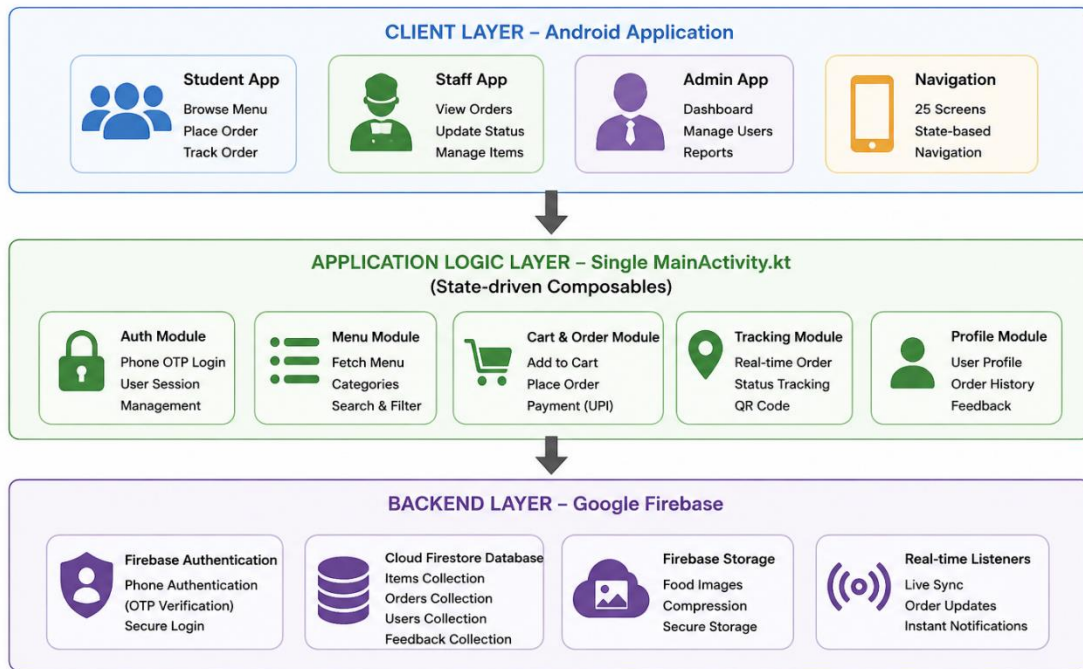
The Application Logic Layer acts as the core processing unit of the system. It is implemented using a single main activity with state-driven composables. This layer consists of multiple modules such as Authentication Module, Menu Module, Cart and Order Module, Tracking Module, and Profile Module. These modules handle business logic like user authentication, menu retrieval, cart operations, order placement, and real-time tracking.

The Backend Layer is powered by Google Firebase services. Firebase Authentication is used for secure OTP-based login. Cloud Firestore is used to store and manage data such as users, menu items, orders, and feedback. Firebase Storage is used for storing food images, and real-time listeners ensure instant updates for order tracking and notifications.

This layered architecture ensures scalability, real-time data synchronization, and efficient separation of concerns, making the system easy to maintain and extend.

SYSTEM ARCHITECTURE DIAGRAM

BVRIT Smart Canteen System



3.4 Algorithms, Flowcharts and Mathematical Formulas

Login Flow (OTP Based)

The User Login Flow Diagram illustrates the step-by-step process of user authentication using phone number and OTP verification.

The process begins when the user launches the application and selects the login option. The user enters their registered mobile number, which is then validated against the Users database. If the phone number does not exist, an error message is displayed.

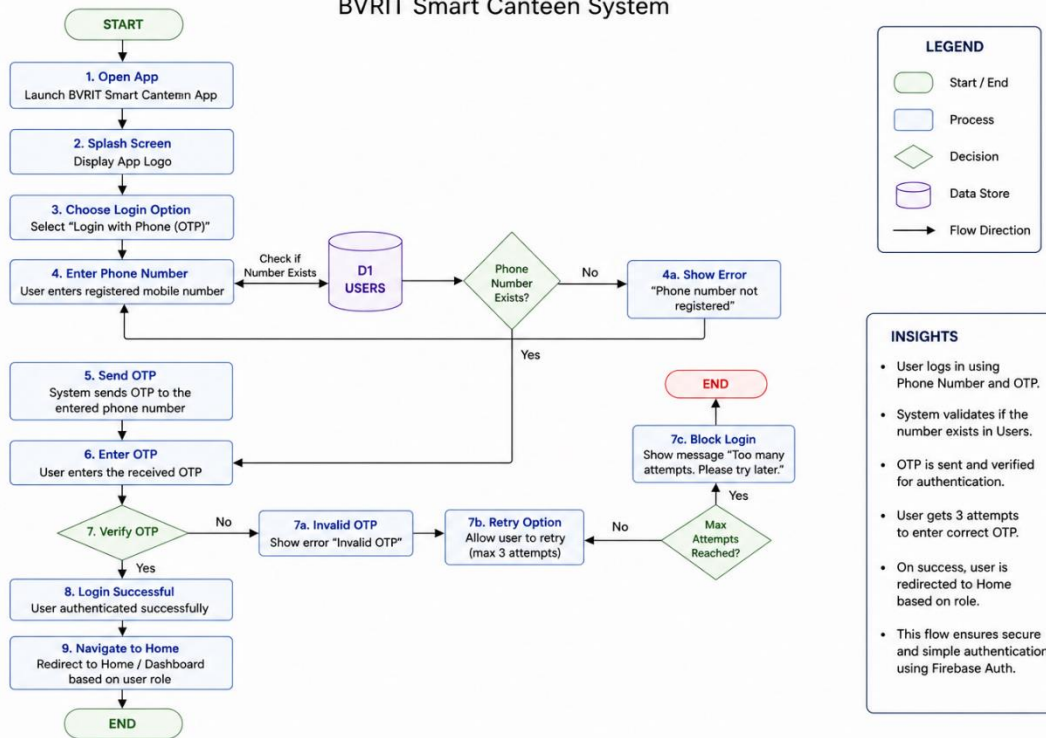
If the number is valid, the system sends a One-Time Password (OTP) to the user's mobile number using Firebase Authentication. The user enters the received OTP, and the system verifies it.

If the OTP is correct, the user is successfully authenticated and redirected to the home screen or dashboard based on their role. If the OTP is incorrect, the user is given up to three attempts to retry. After exceeding the maximum attempts, the login process is blocked temporarily.

This flow ensures secure, simple, and efficient authentication, eliminating the need for passwords while maintaining data security.

USER LOGIN FLOW (FLOWCHART)

BVRIT Smart Canteen System



Order Placement Flow

The Order Placement Flow Diagram represents the complete process of ordering food in the Smart Canteen System, starting from app usage to order collection.

The process begins when the user opens the application and logs in using OTP authentication. After login, the user selects the canteen location (D2 or Food Court) and chooses the dining mode (Dine-in or Take-away).

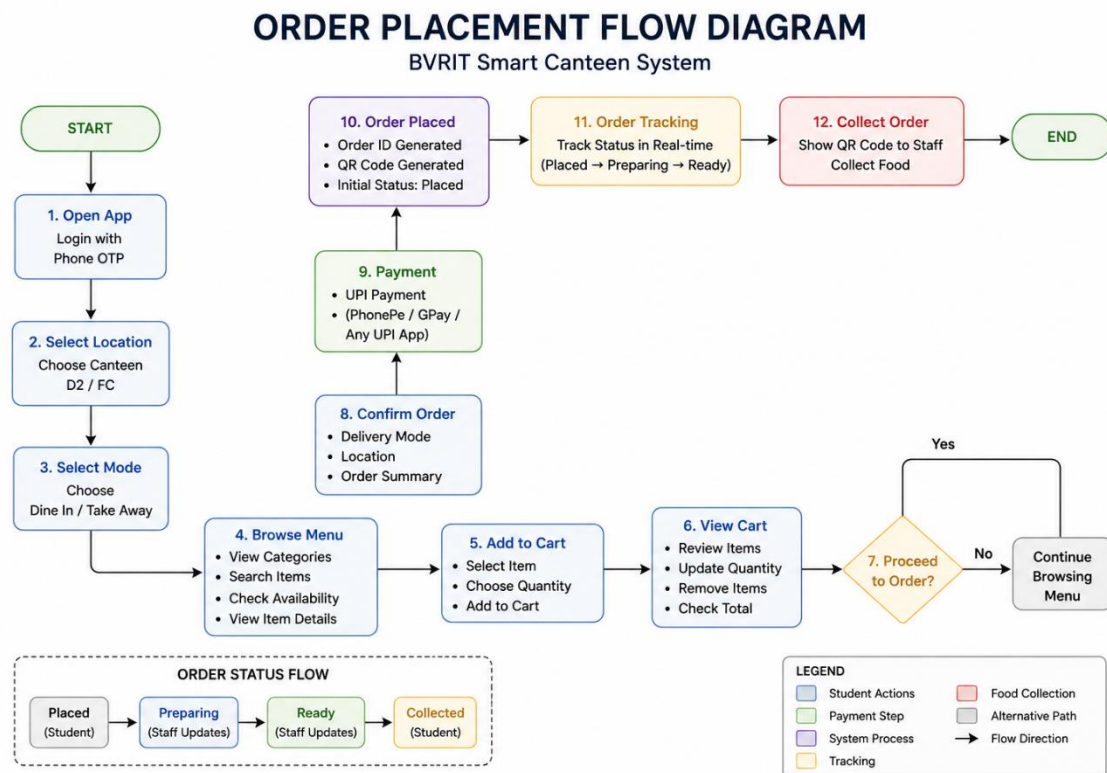
Next, the user browses the menu, searches for items, checks availability, and adds desired items to the cart. In the cart section, the user can review items, update quantities, or remove items. The system calculates the total amount dynamically.

Once satisfied, the user proceeds to confirm the order by reviewing details such as location, mode, and order summary. After confirmation, the user makes payment using UPI or other supported methods.

After successful payment, the order is placed, and a unique Order ID along with a QR code is generated. The order status is initially set to "Placed".

The user can then track the order in real time, where the status changes from Placed → Preparing → Ready as updated by the staff. Finally, the user collects the order by showing the QR code to the staff, completing the process.

This flow ensures a smooth, queue-free, and efficient food ordering experience.



Order Total Calculation

$$Total = \sum(price_i \times quantity_i)$$

- Total = Sum of (item price × quantity)

CHAPTER 4

DESIGN

CHAPTER 4

DESIGN

4.1 Modules Introduction

The Smart Canteen Pre-Ordering System is designed using a modular architecture, where each module is responsible for a specific functionality of the application. This modular approach improves system organization, simplifies development, and makes future enhancements easier. Each module interacts with Firebase services to ensure real-time data synchronization and efficient operation.

4.1.1 Authentication Module

The Authentication Module manages all user identity operations within the application. Unlike traditional systems that use email and password, this system uses Firebase Phone Authentication with OTP verification for secure and quick login.

When the user enters their mobile number, an OTP is sent via Firebase. Upon successful verification, the system checks whether the user already exists in the database. If it is a first-time login, a new user document is created in Firestore with basic details such as phone number and user ID.

This module also maintains session persistence, allowing users to remain logged in unless they explicitly log out. It ensures secure access control across all features of the application.

4.1.2 Menu Module

The Menu Module is responsible for displaying food items available in both canteens — Daffodils 2 (D2) and Food Court (FC). It retrieves data from the Firebase Firestore database in real time.

The module filters items based on:

- Selected canteen location (D2 or FC)
- Availability status

It supports search functionality and category-based filtering, allowing users to easily find desired items. The items are displayed in a structured grid layout, ensuring a clean and user-friendly interface.

Additionally, any updates made by admin (such as price changes or availability) are instantly reflected on the user interface without requiring a refresh, thanks to Firestore's real-time listeners.

4.1.3 Cart Module

The Cart Module manages the user's selected food items before order placement. It acts as a temporary storage area where users can review and modify their selections.

Users can:

- Add items to the cart
- Increase or decrease item quantity
- Remove items
- Add special notes for items

The module dynamically calculates the total price as items are modified. The cart state is maintained using in-memory state (Jetpack Compose state management), **ensuring smooth performance and instant updates.**

4.1.4 Order Management Module

The Order Management Module handles the complete process of order creation and storage. When a user confirms their order, this module generates an order object containing all necessary details such as items, quantities, total amount, selected canteen, and dining mode.

This data is stored in the Firestore "orders" collection, where each order is uniquely identified. After successful order placement, the cart is cleared, and the user is redirected to the order confirmation screen.

This module ensures that all order data is securely stored and easily retrievable for tracking and history purposes.

4.1.5 Order Tracking Module

The Order Tracking Module provides real-time updates on the status of placed orders. It uses Firestore snapshot listeners to continuously monitor changes in the order document.

The order progresses through the following stages:

- Placed
- Preparing
- Ready

Whenever the status is updated by staff, the change is immediately reflected on the user's screen. The module also includes a countdown timer and visual indicators to improve user experience and provide clarity about order readiness.

4.1.6 Staff Module

The Staff Module is designed for canteen staff to manage ongoing operations efficiently. It displays all active orders for a specific canteen in real time.

Staff members can:

- View incoming orders
- Update order status
- Monitor order details
- Manage item availability

This module ensures smooth communication between kitchen staff and users, reducing delays and improving service efficiency.

4.1.7 Admin Module

The Admin Module provides advanced control over the system. It allows administrators to manage menu items and monitor system activity.

Admin functionalities include:

- Adding new food items
- Updating item details (price, cooking time, availability)
- Deleting items
- Uploading images to Firebase Storage
- Viewing user feedback

Each admin is associated with a specific canteen (D2 or FC), ensuring proper management and data segregation.

4.1.8 Profile Module

The Profile Module allows users to view and update their personal information stored in Firestore.

Users can:

- View name and phone number
- Update name and roll number
- Access order history
- Logout from the system

This module ensures that user data remains consistent and up-to-date.

4.2 DFD / ER / UML Diagrams

4.2.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) models the flow of information within the system. It shows how data moves between processes, data stores, and external entities without describing implementation details.

LEVEL 0 - CONTEXT DIAGRAM (DFD)

The Level 0 Data Flow Diagram (Context Diagram) provides a high-level overview of the Smart Canteen System, showing the interaction between external entities and the system as a whole.

The system is represented as a single process called "BVRIT Smart Canteen System". It interacts with three main external entities: Student, Staff, and Admin.

The Student sends requests such as login, browsing menu, placing orders, making payments, tracking orders, and giving feedback. In response, the system provides menu details, order status, payment status, notifications, and order history.

The Staff interacts with the system to view incoming orders, update order status, scan QR codes, and manage menu items. The system provides order details and new order notifications to the staff.

The Admin manages users, menu items, feedback, and reports. The system sends reports, user data, and feedback details to the admin.

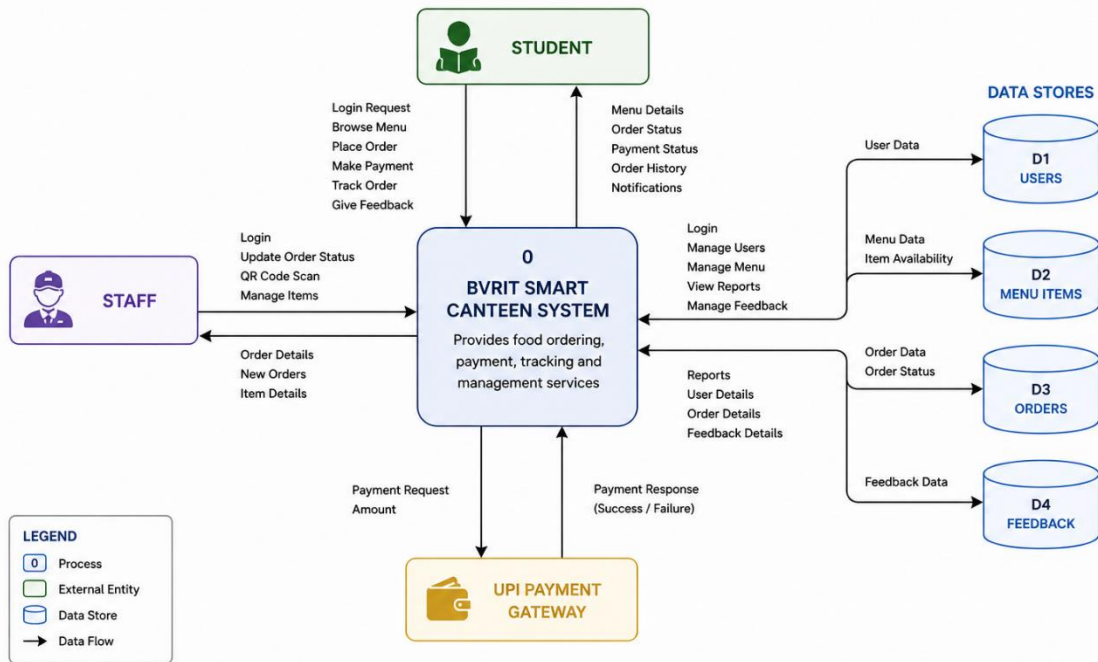
The system also interacts with data stores such as Users, Menu Items, Orders, and Feedback, where all relevant data is stored and retrieved.

Additionally, an external Payment Gateway is used to process payment requests and return payment responses.

This diagram gives a *simplified overview of system boundaries and data interactions.

DATA FLOW DIAGRAM (DFD) – LEVEL 0

BVRIT Smart Canteen System



LEVEL 1 - EXPANDED PROCESSES (DFD)

The Level 1 Data Flow Diagram provides a detailed breakdown of the internal processes of the system by expanding the Level 0 diagram into multiple functional processes.

The system is divided into seven main processes:

1.0 User Authentication

Handles user login and registration using OTP verification. It interacts with the Users data store to validate and store user information.

2.0 Menu Management

Retrieves and manages menu items from the Menu Items data store. It provides updated menu details to users.

3.0 Cart Management

Manages adding, updating, and removing items in the cart. It temporarily stores cart data before order placement.

4.0 Order Management

Handles order creation, stores order details in the Orders data store, and manages order status updates.

5.0 Payment Processing

Processes payment requests through an external Payment Gateway and stores transaction details in the Payments data store.

6.0 Order Tracking

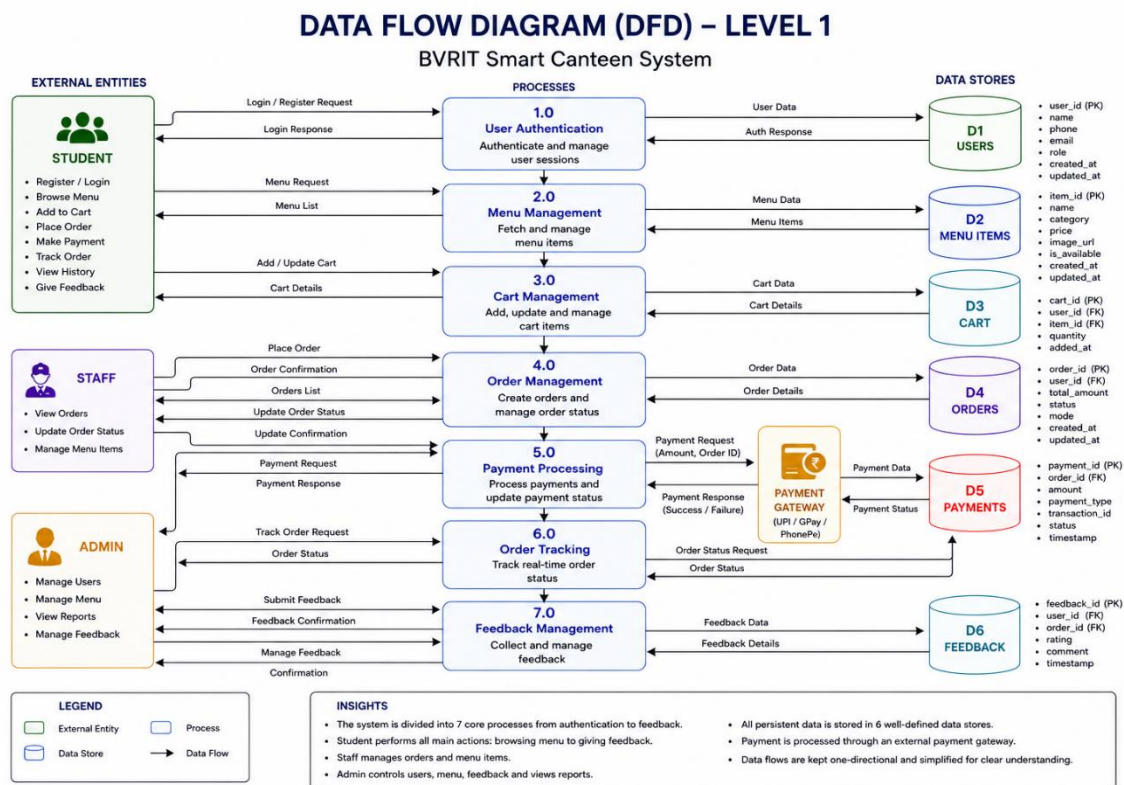
Provides real-time order status updates by retrieving data from the Orders data store.

7.0 Feedback Management

Collects and stores user feedback in the Feedback data store.

Each process interacts with specific data stores (Users, Menu Items, Cart, Orders, Payments, Feedback), ensuring proper data flow and separation of functionality.

This diagram gives a detailed view of how data moves inside the system and how each module functions*.



4.2.2 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) represents the database structure of the BVRIT Smart Canteen System and shows how different entities are related to each other.

The main entities in the system include Users, Profiles, Orders, Order_Items, Menu_Items, Canteens, and Staffs.

The Users entity stores details such as user ID, name, phone number, email, and role (student, staff, admin). Each user has exactly one Profile, which contains additional information like address, date of birth, and profile image. This represents a one-to-one relationship.

A user can place multiple Orders, forming a one-to-many relationship. Each order contains details such as order ID, total amount, status, payment status, and mode (dine-in or take-away).

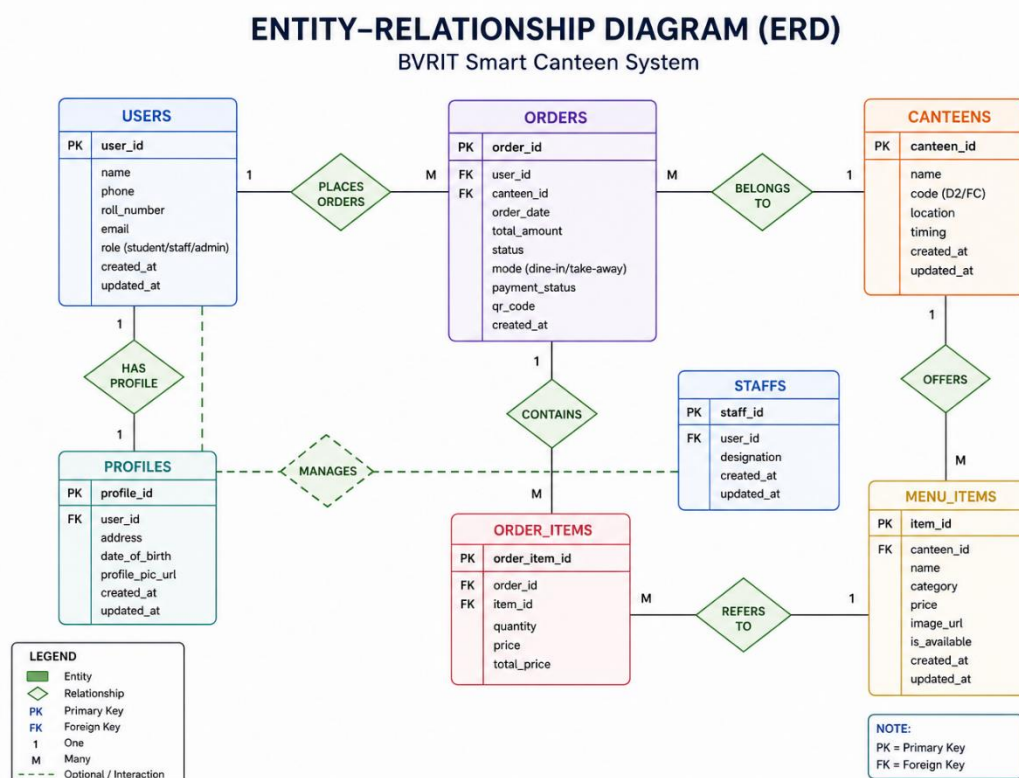
Each order is composed of multiple Order_Items, which represent individual items in the order. This creates a one-to-many relationship between Orders and Order_Items.

The Menu_Items entity stores information about food items such as name, category, price, and availability. Each order item refers to a specific menu item, forming a many-to-one relationship.

The Canteens entity represents different canteen locations, and each canteen offers multiple menu items, creating a one-to-many relationship.

The Staffs entity represents staff members associated with the system, who are responsible for managing orders and menu items.

Overall, the ER diagram ensures proper data organization, normalization, and relationship mapping, enabling efficient storage and retrieval of information.



4.2.3 UML Diagrams

The Use Case Diagram illustrates the functional requirements of the Smart Canteen System by showing the interactions between users and the system.

There are three primary actors in the system: Student, Staff, and Admin, along with an external Payment Gateway.

The Student is the main user of the system and performs actions such as registering or logging in, browsing the menu, searching and filtering items, adding items to the cart, placing orders, making payments, tracking order status, and giving feedback.

The Staff interacts with the system to view incoming orders, update order status (such as preparing or ready), and manage menu items.

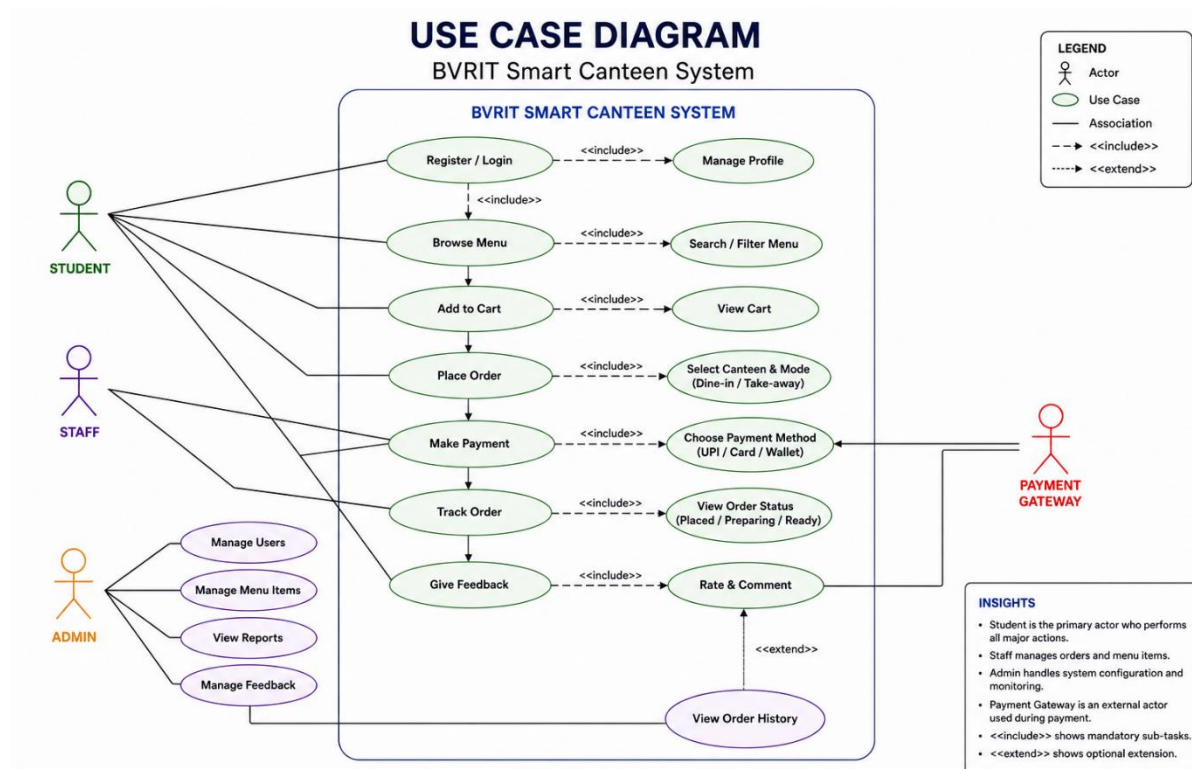
The Admin is responsible for higher-level management tasks such as managing users, managing menu items, viewing reports, and managing feedback.

The Payment Gateway acts as an external actor that processes payment transactions during the order process.

The diagram also includes relationships such as <<include>>, which represents mandatory actions (e.g., placing an order includes selecting canteen and payment method), and <<extend>>, which represents optional actions (e.g., rating and commenting after viewing order history).

This diagram provides a clear understanding of system functionality and user interactions.

USE CASE DIAGRAM



CLASS DIAGRAM

The Class Diagram represents the object-oriented structure of the Smart Canteen System, showing classes, their attributes, methods, and relationships.

The main classes include User, Profile, Cart, CartItem, Order, OrderItem, MenuItem, Canteen, Payment, Notification, Feedback, and Payment Gateway.

The User class contains attributes such as user ID, name, phone, email, and role, and provides methods for login and profile updates. Each user is associated with one Profile.

The Cart class holds items selected by the user, and it contains multiple CartItem objects. Each cart item includes item details, quantity, and price.

When an order is placed, a Cart is converted into an Order, which contains multiple OrderItem objects. The Order class manages order details such as status, total amount, and mode.

The MenuItem class represents food items available in the system, and it is linked to the Canteen class, which represents the location offering those items.

The Payment class handles transaction details and interacts with an external *Payment Gateway to process payments.

The Notification class is used to send updates to users regarding order status changes.

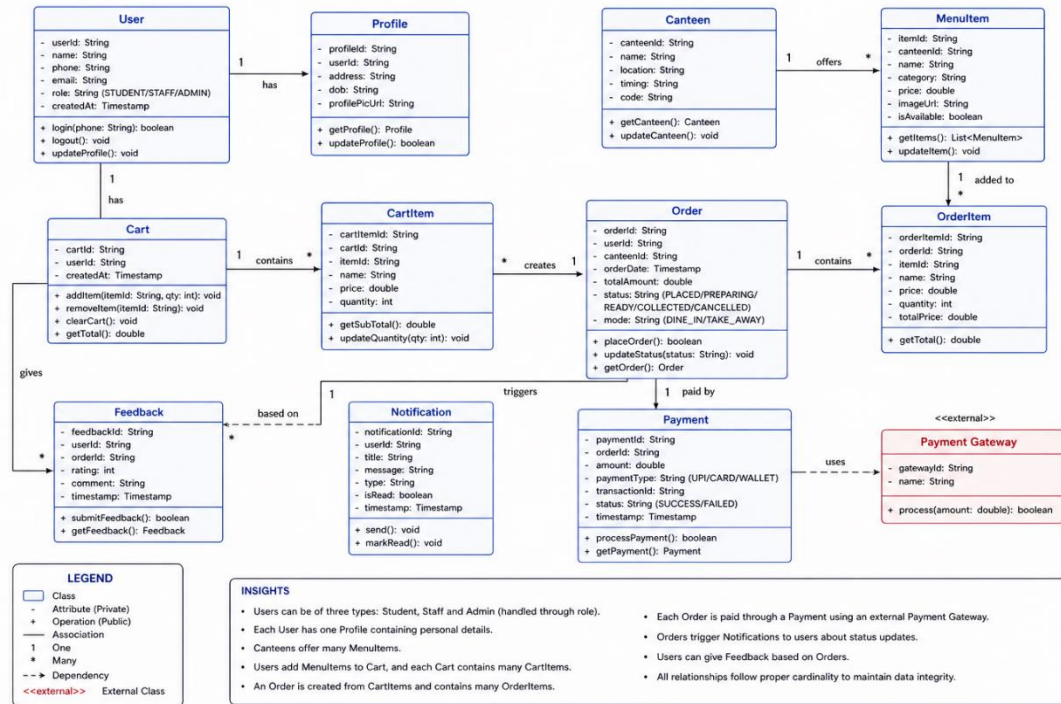
The Feedback class allows users to submit ratings and comments based on their orders.

The relationships between classes include one-to-one, one-to-many, and dependency relationships, ensuring proper system design.

This diagram provides a clear blueprint for implementation and system design using object-oriented principles.

CLASS DIAGRAM

BVRIT Smart Canteen System



4.3 Module Design and Organization

The application follows the MVVM (Model-View-ViewModel) architectural pattern as recommended by Google for Android development. The project package structure is organized as follows:

PROJECT PACKAGE STRUCTURE

com.bvrit.canteen/

└─ ui/

| └─ auth/ (LoginActivity, RegisterActivity, AuthViewModel)

| └─ home/ (HomeFragment, HomeViewModel)

| └─ menu/ (MenuFragment, MenuItemAdapter, MenuViewModel)

| └─ cart/ (CartFragment, CartViewModel)

| └─ order/ (OrderFragment, OrderHistoryFragment, OrderViewModel)

| └─ payment/ (PaymentActivity, PaymentViewModel)

| └─ admin/ (AdminDashboardFragment, AdminOrderFragment, AdminMenuFragment)

└─ data/

| └─ model/ (User.kt, Order.kt, MenuItem.kt, OrderItem.kt)

| └─ repository/ (AuthRepository.kt, OrderRepository.kt, MenuRepository.kt)

| └─ remote/ (FirebaseService.kt, FCMSERVICE.kt)

└─ utils/ (Extensions.kt, Constants.kt, NetworkUtils.kt)

└─ di/ (AppModule.kt, NetworkModule.kt)

CHAPTER 5

OUTPUT SCREENSHOTS AND DESCRIPTION

CHAPTER 5

OUTPUT SCREENSHOTS AND DESCRIPTION

This chapter presents the key screens of the Smart Canteen Android application, along with descriptions of each screen's functionality and design rationale. The screens described below represent the implemented and tested modules of the system.

5.1 Splash / Welcome Screen

The Splash Screen serves as the branded entry point of the application. It displays a full-screen blue patterned background along with the text "BVRIT" in large white font and "Smart Canteen" as a subtitle. The text elements fade in using smooth animations.

The screen is displayed for approximately 2.7 seconds while the application initializes. During this time, Firebase Authentication checks whether a user session already exists. If an active session is found, the user is directly navigated to the Location Selection Screen; otherwise, the user is redirected to the Phone Login Screen. The screen contains no interactive elements and ensures a visually engaging startup experience.

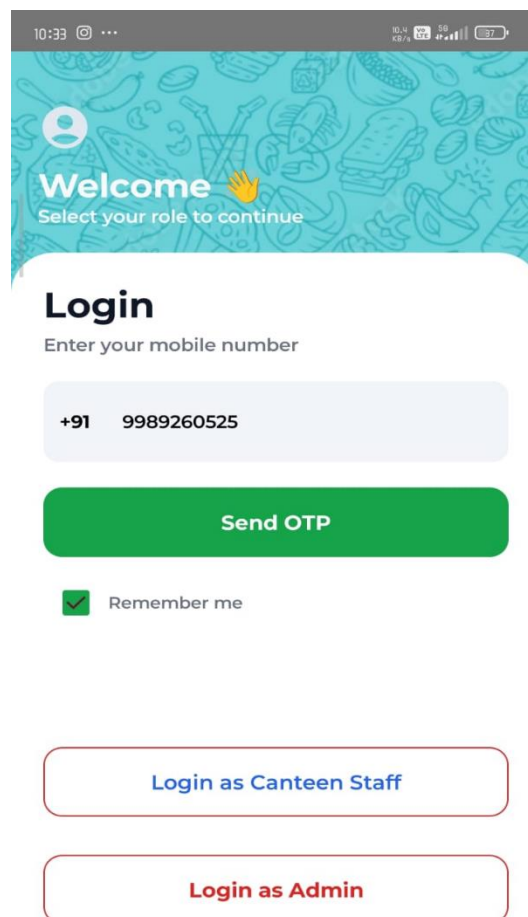


5.2 Phone Login Screen

The Phone Login Screen acts as the primary authentication entry point for students. It collects the user's 10-digit Indian mobile number and initiates OTP-based authentication using Firebase.

The screen features a split layout with a blue patterned top section and a white card at the bottom. The card includes a phone number input field prefixed with "+91", a "Send OTP" button, a "Remember Me" checkbox, and options for Staff Login and Admin Login.

Upon clicking "Send OTP", the entered number is validated and Firebase Phone Authentication is triggered. If successful, the user is navigated to the OTP Verification Screen along with the verification ID.



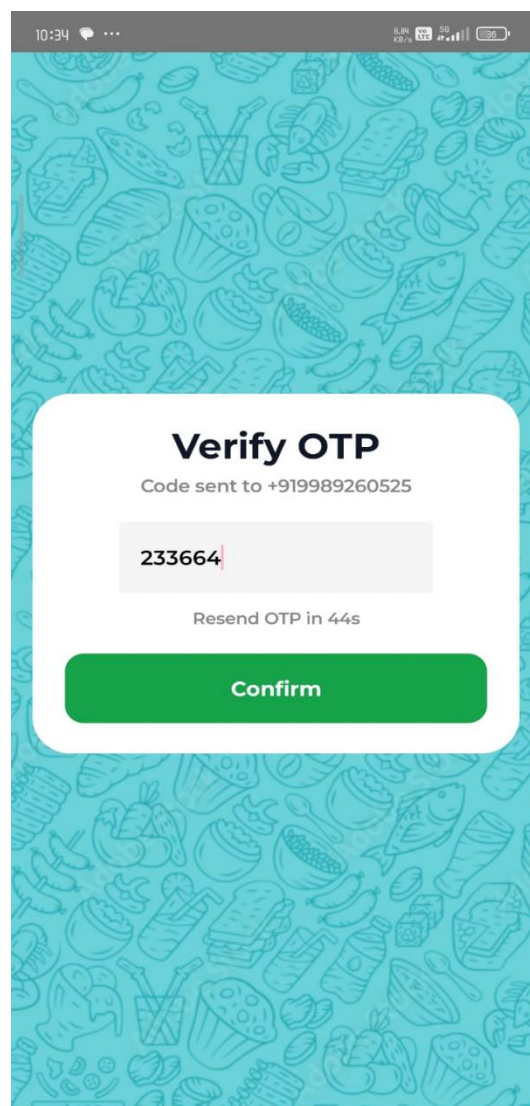
The screenshot shows a mobile application interface. At the top, there is a status bar with the time 10:33, signal strength, and battery level. Below the status bar is a blue header with a pattern of food icons. The header contains a white user icon, the text "Welcome 🙌", and "Select your role to continue". Below the header is a white card with the title "Login" and the instruction "Enter your mobile number". The card features a light blue input field with the text "+91 9989260525". Below the input field is a green button labeled "Send OTP". Underneath the button is a checkbox labeled "Remember me" which is checked. At the bottom of the card are two buttons: "Login as Canteen Staff" and "Login as Admin".

5.3 OTP Verification Screen

The OTP Verification Screen is used to complete user authentication by verifying the OTP sent via SMS.

The screen includes a 6-digit OTP input field, a countdown timer of 60 seconds, and a "Confirm" button. After the timer expires, a "Resend OTP" option becomes available.

On successful OTP verification, Firebase authenticates the user. If it is the first login, a new user record is created in Firestore with basic details such as phone number and timestamp. The user is then navigated to the Location Selection Screen.

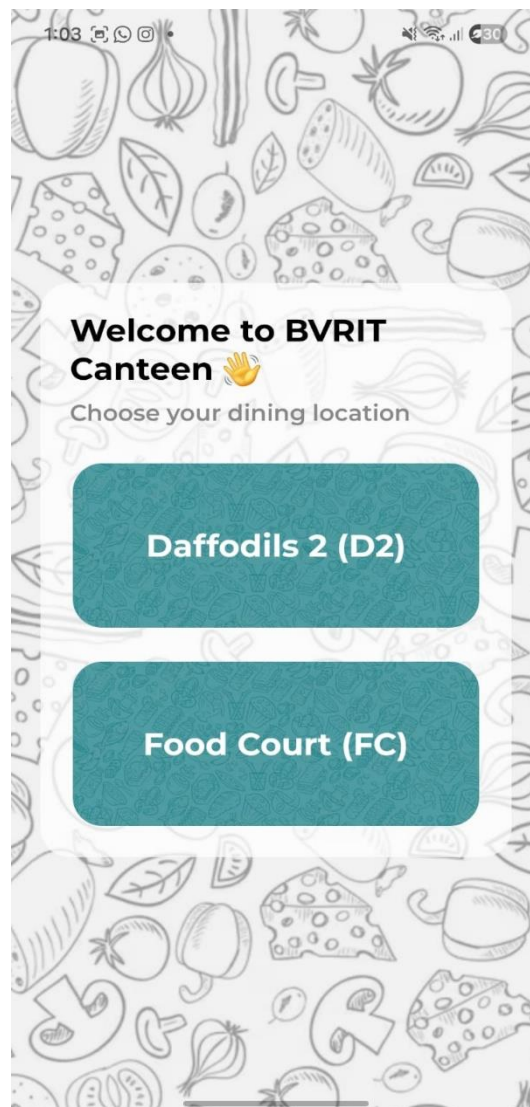


5.4 Location Selection Screen

This screen allows users to choose their preferred canteen location for ordering.

The interface displays two selectable cards: "Daffodils 2 (D2)" and "Food Court (FC)". Each card is visually styled with a blue patterned background and clear text labels.

When a user selects a location, it is stored as part of the session state and used throughout the ordering process. The user is then navigated to the Mode Selection Screen.

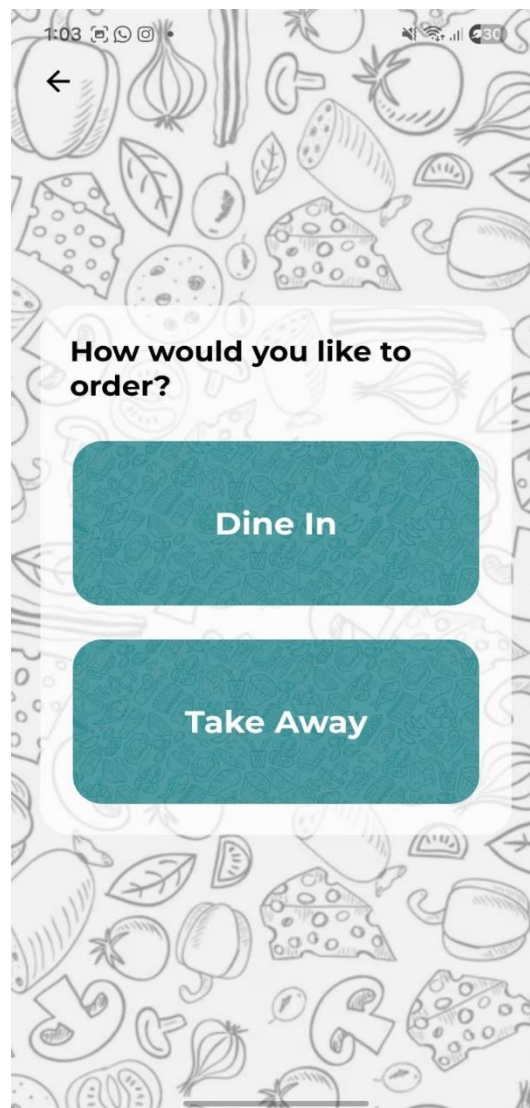


5.5 Mode Selection Screen

The Mode Selection Screen captures whether the user intends to dine at the canteen or take the order away.

The screen presents two options: "Dine In" and "Take Away", each represented with icons and styled cards.

Selecting a mode stores the preference and navigates the user to the Home Screen. This mode information is later associated with each order.

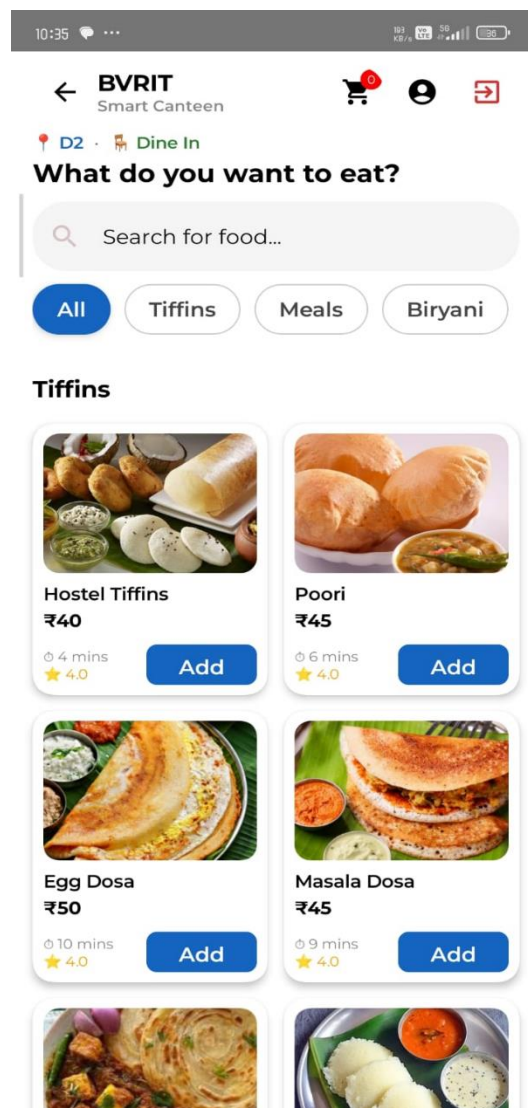


5.6 Home Screen — D2 Menu Display

The Home Screen for Daffodils 2 (D2) allows users to browse available food items

. It includes a sticky header with navigation icons, location (D2), dining mode, a search bar, and category filters such as Tiffin's, Meals, and Snacks. Items are displayed in a two-column grid layout.

Data is fetched in real time from Fire store, showing only available items for D2. The menu uses time-based category prioritization. Users can search, filter items, and view live order status through a floating banner.

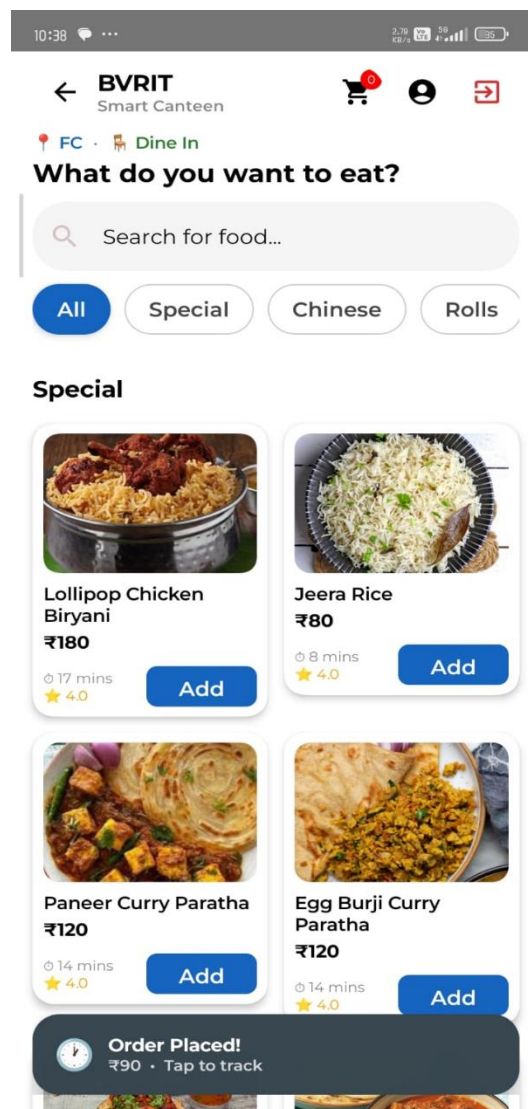


5.7 Home Screen — FC Menu Display

The Home Screen for the Food Court (FC) displays a variety of food items available in the FC canteen.

It features a sticky header with navigation icons, location (FC), dining mode, a search bar, and category filters such as Fast Food, Snacks, and Beverages. Items are shown in a two-column grid layout.

Data is loaded in real time from Firestore, showing only available items for FC. Unlike D2, categories are displayed as defined without time-based prioritization. Users can search, filter items, and track active orders through a floating status banner.

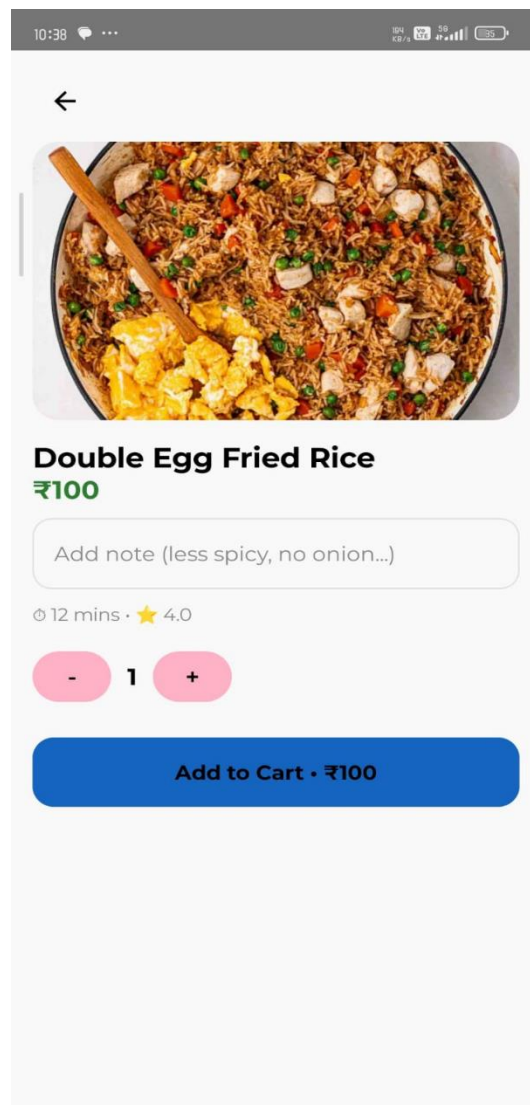


5.8 Item Detail Screen

The Item Detail Screen provides complete information about a selected food item.

It displays the item image, name, price, cooking time, rating, and a notes field for customization. A quantity selector allows users to adjust the number of items.

The "Add to Cart" button dynamically updates the total price. If the same item with identical notes already exists in the cart, the system intelligently updates the quantity instead of duplicating entries.

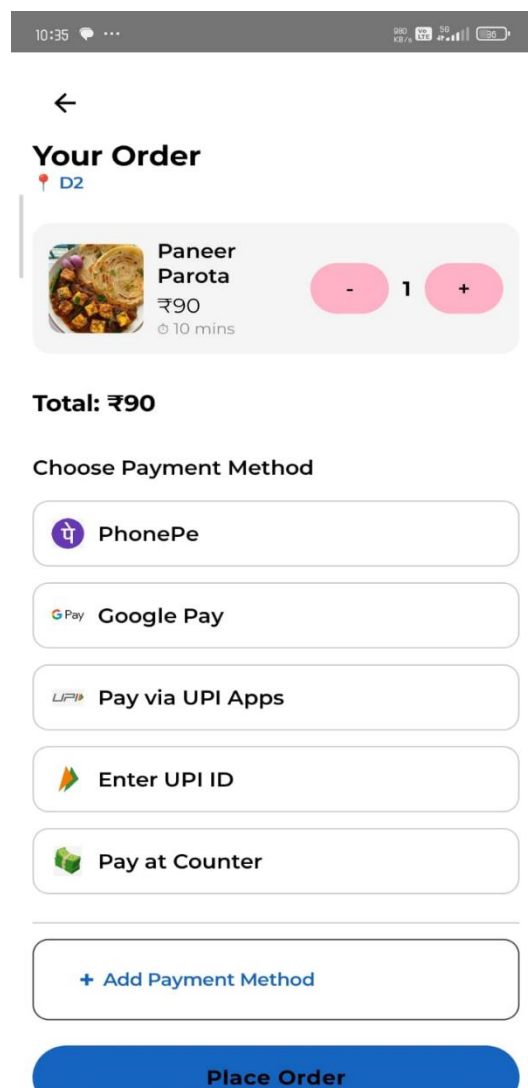


5.9 Cart Screen — D2 (Daffodils 2)

The Cart Screen for D2 displays all selected food items before placing an order.

It shows item details such as name, price, quantity controls, and notes, along with the total amount.

Users can modify or remove items and select a payment method. On placing the order, the details are saved and the user is navigated to the Order Confirmation screen.

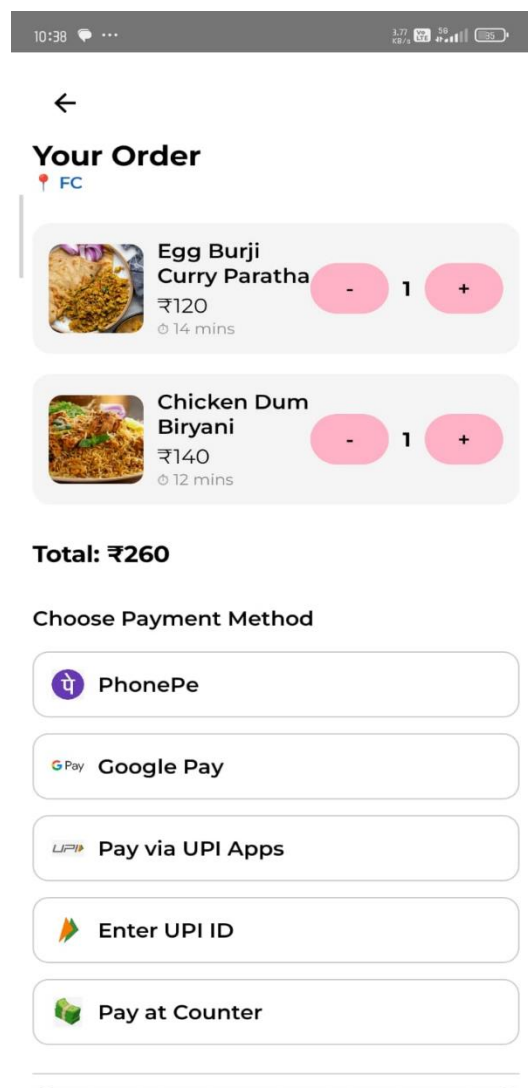


5.10 Cart Screen — FC (Food Court)

The Cart Screen for FC shows all selected items from the Food Court menu.

It includes item details, quantity controls, and total amount, along with payment method selection.

Users can update the cart and place the order, after which the order is saved and the user is directed to the Order Confirmation screen.

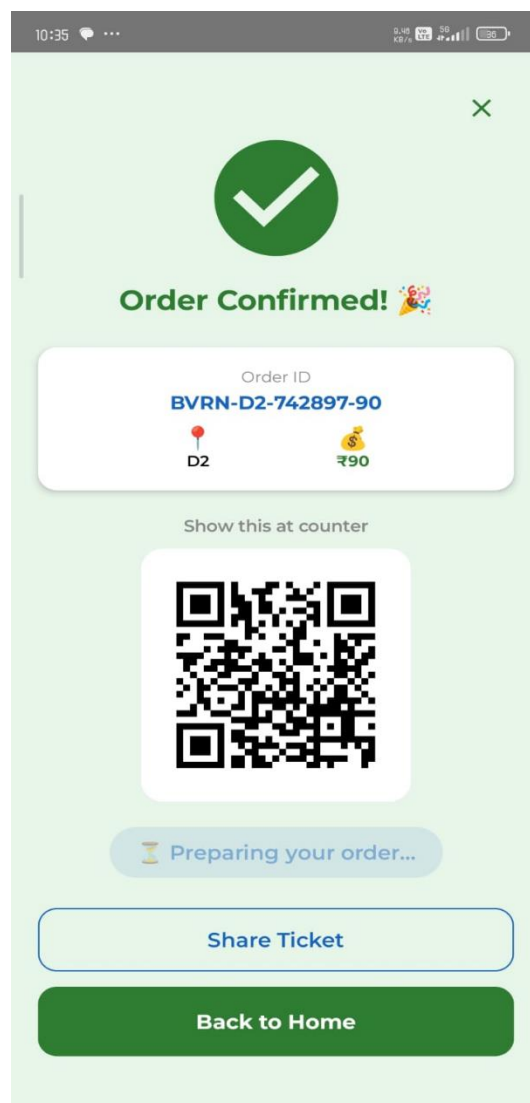


5.11 Order Confirmation Screen — D2 (Daffodils 2)

The Order Confirmation Screen appears after a successful order placement.

It displays a confirmation message along with the Order ID, total amount, and selected canteen (D2).

A QR code is generated for verification at the D2 counter and can be shared via external apps. Users can either return to the Home Screen or proceed to order tracking.

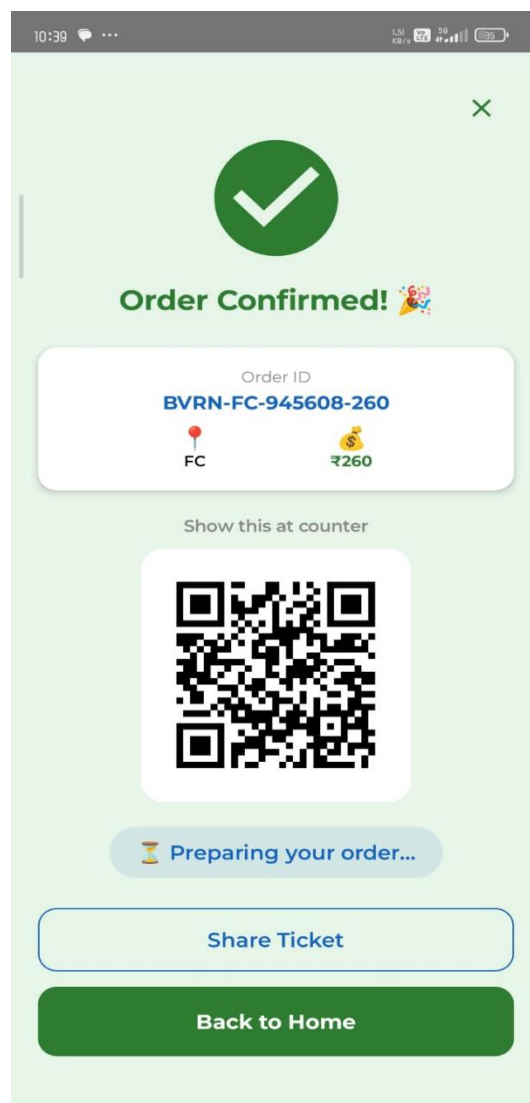


5.12 Order Confirmation Screen — FC (Food Court)

The Order Confirmation Screen appears after a successful order placement.

It displays a confirmation message along with the Order ID, total amount, and selected canteen (FC).

A QR code is generated for verification at the FC counter and can be shared via external apps. Users can either return to the Home Screen or proceed to order tracking.

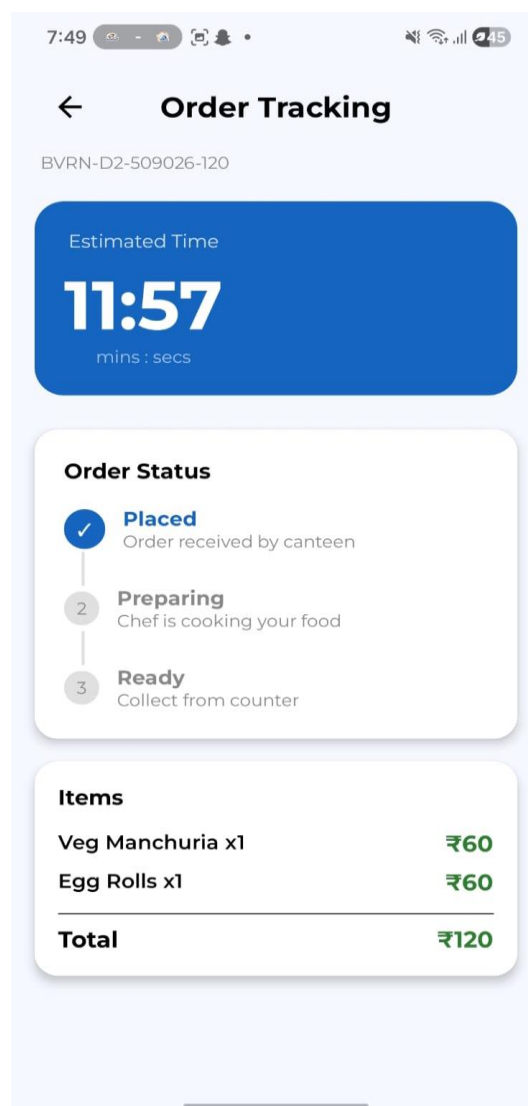


5.13 Order Tracking Screen

The Order Tracking Screen provides real-time updates on the order status.

It visually represents three stages: Placed, Preparing, and Ready, using a progress indicator. A countdown timer shows the estimated preparation time.

The screen uses Firestore listeners to update status instantly. Once the order is marked "Ready", the timer stops and the user can proceed to collect the order.

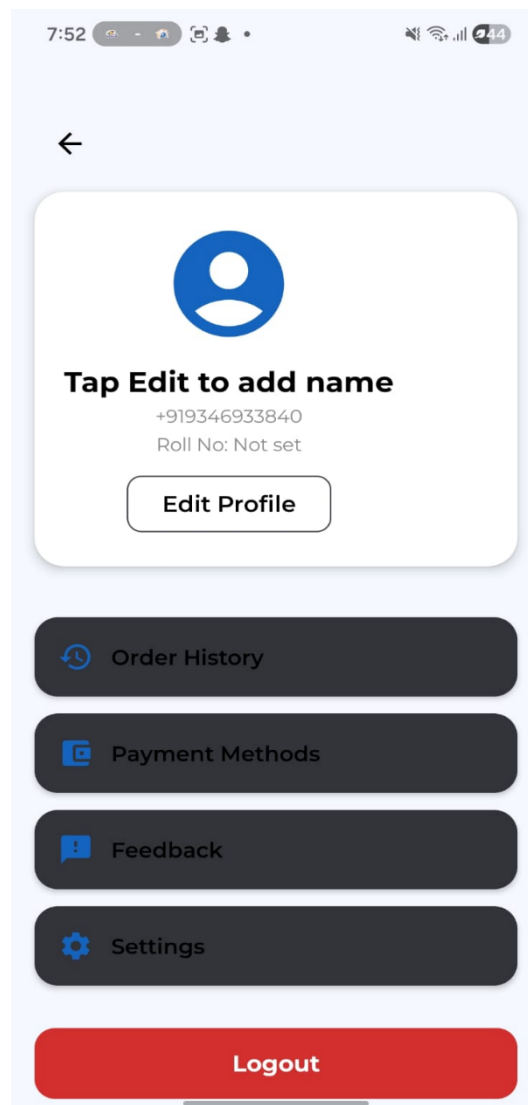


5.14 Profile Screen

The Profile Screen displays and allows editing of user information such as name, phone number, and roll number.

Users can toggle into edit mode to update their details, which are then saved to Firestore.

Additional options such as Order History, Feedback, and Settings are available, though only Order History is fully functional. A Logout option is also provided.

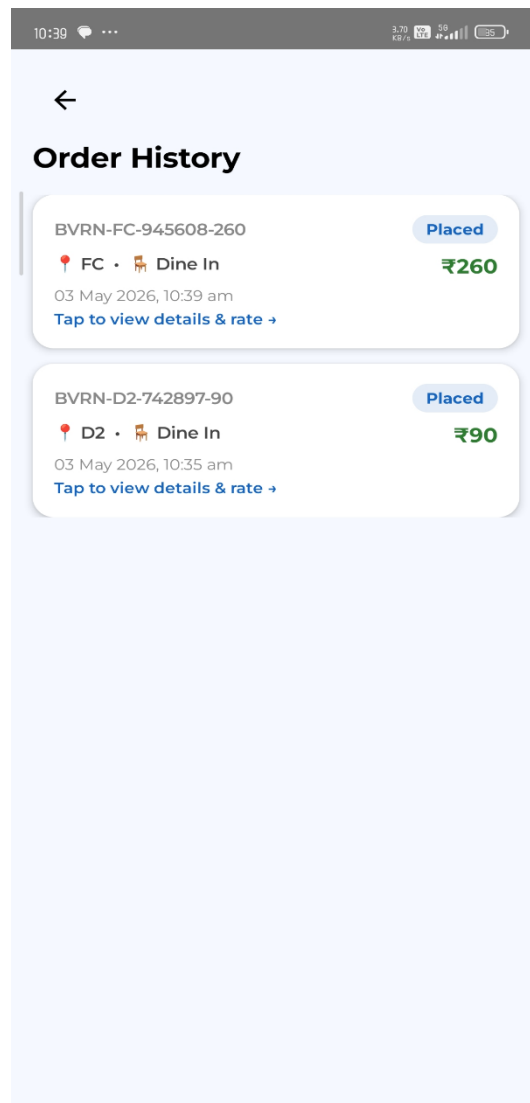


5.15 Order History Screen

The Order History Screen lists all previous orders made by the user in reverse chronological order.

Each order card displays order ID, location, mode, total amount, timestamp, and status.

Selecting an order navigates to the Order Detail Screen for further information.



5.16 Order Detail Screen — D2 (Daffodils 2)

This screen shows detailed information of a past D2 order.

It includes the item list, payment method, total amount, and order status.

Users can submit feedback using star rating and comments. Once submitted, it cannot be edited.

The screenshot shows a mobile application interface for viewing order details. At the top, there is a status bar with the time 10:36, signal strength, and battery level. Below the status bar is a back arrow icon. The main heading is "Order Details" with the order ID "BVRN-D2-742897-90" below it. A table-like structure displays the following information: Location (D2), Mode (Dine In), Payment (Pay at Counter), and Status (Placed). Below this, the "Items Ordered" section lists "Paneer Parota" with a quantity of "x1" and a price of "₹90". The total amount is displayed as "Total: ₹90". At the bottom, there is a "Rate this Order" section with five star icons for rating. Below the stars is a text input field labeled "Share your experience...". A blue button labeled "Submit Feedback" is positioned at the bottom of the feedback section.

Location	D2
Mode	Dine In
Payment	Pay at Counter
Status	Placed

Items Ordered

Item	Quantity	Price
Paneer Parota	x1	₹90

Total: ₹90

Rate this Order

☆☆☆☆☆

Share your experience...

Submit Feedback

5.17 Order Detail Screen — FC (Food Court)

This screen displays complete details of a past FC order.

It shows the item list, payment method, total amount, and order status.

Users can provide feedback through rating and comments, which cannot be modified after submission.

The screenshot shows a mobile application interface for viewing order details. At the top, there is a status bar with the time 11:38, a back arrow, and network/battery icons. Below the status bar is a back arrow and the title "Order Details" with the order ID "BVRN-FC-485274-260". A table displays order details: Location (FC), Mode (Dine In), Payment (Pay at Counter), and Status (Placed). Below this is a section titled "Items Ordered" listing "Egg Burji Curry Paratha" (x1 ₹120) and "Chicken Dum Biryani" (x1 ₹140), with a total of ₹260. At the bottom, there is a "Rate this Order" section with five stars, a text input field for feedback, and a "Submit Feedback" button.

Location	FC
Mode	Dine In
Payment	Pay at Counter
Status	Placed

Items Ordered

Egg Burji Curry Paratha	x1 ₹120
Chicken Dum Biryani	x1 ₹140
Total: ₹260	

Rate this Order

☆☆☆☆☆

Share your experience...

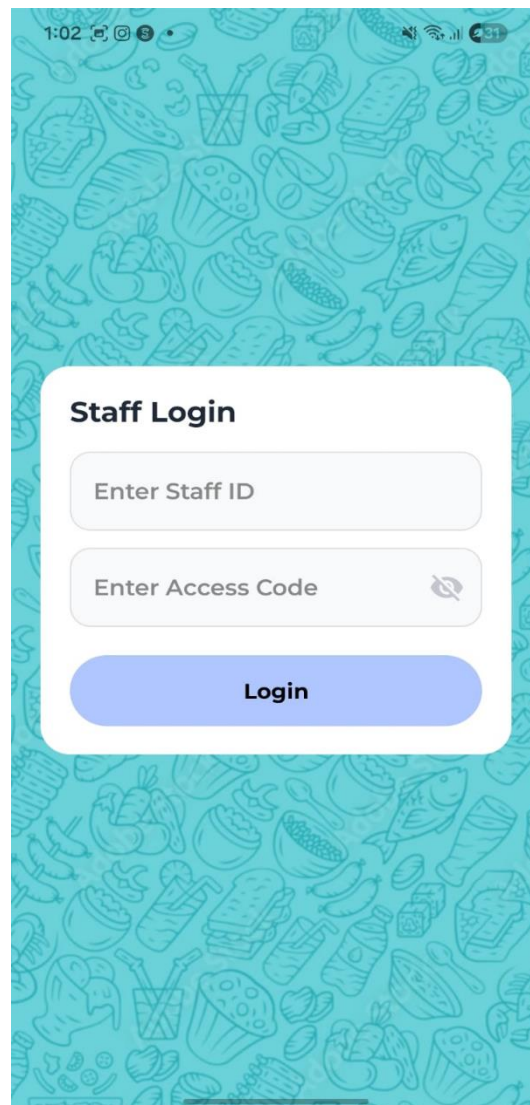
Submit Feedback

5.18 Staff Login Screen

The Staff Login Screen is used by canteen staff to access operational features.

It contains fields for Staff ID and password. Credentials are validated against predefined data.

Upon successful login, the user is navigated to the Staff Home Screen with location-specific access.

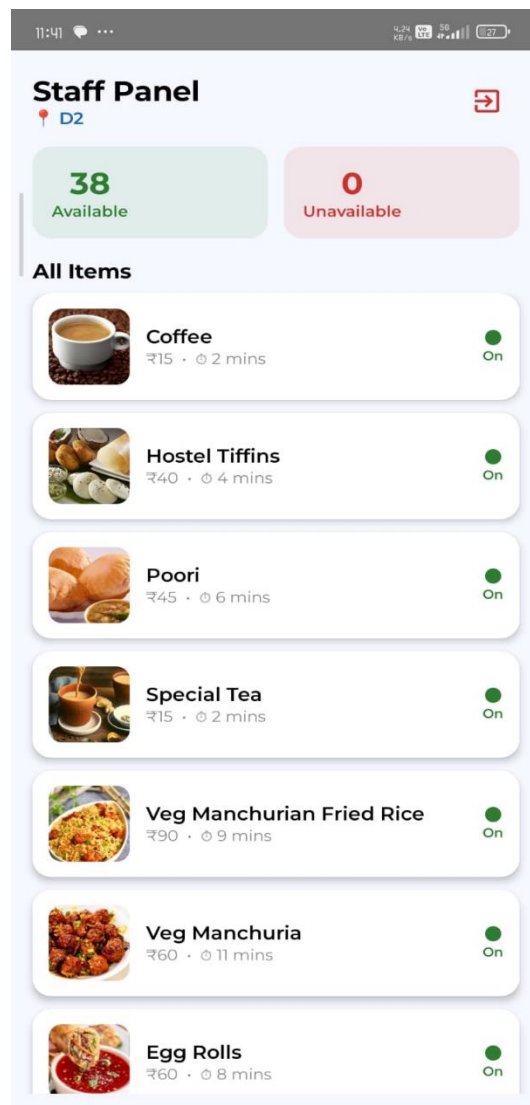
The image shows a mobile application interface for a staff login screen. The background is a teal color with a repeating pattern of various food items like sandwiches, fruits, and drinks. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle, the text "Staff Login" is displayed in bold. Below it, there are two input fields: the first is labeled "Enter Staff ID" and the second is labeled "Enter Access Code" with an eye icon to its right, indicating a password field. At the bottom of the white rectangle is a blue rounded button with the text "Login" in white. The top of the screen shows a status bar with the time "1:02", signal strength, and battery level "43%".

5.19 Staff Home Screen — D2 (Daffodils 2)

The Staff Home Screen for D2 is used to manage orders and menu items.

It displays active orders with status controls and a list of all items.

Staff can update order status and manage item availability in real time.

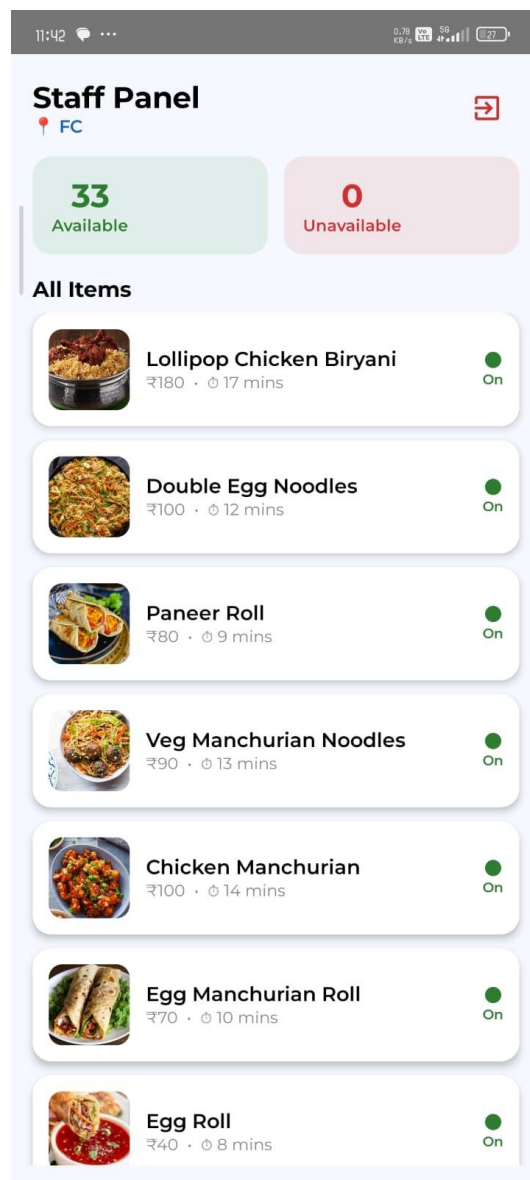


5.20 Staff Home Screen — FC (Food Court)

The Staff Home Screen for FC provides tools to manage orders and menu items.

It shows active orders with status controls and the complete item list.

Staff can update order status and control item availability with real-time updates.



5.21 Staff Item Control Screen

This screen allows staff to manage individual menu items.

Staff can toggle availability and update cooking time.

Changes are saved to Firestore and reflected immediately on the user side.

11:42

58.1 KB/s 5G

27

←



Lollipop Chicken Biryani

₹180

Cooking Time (mins)

17

Availability

☒ Available ☐ Not Available

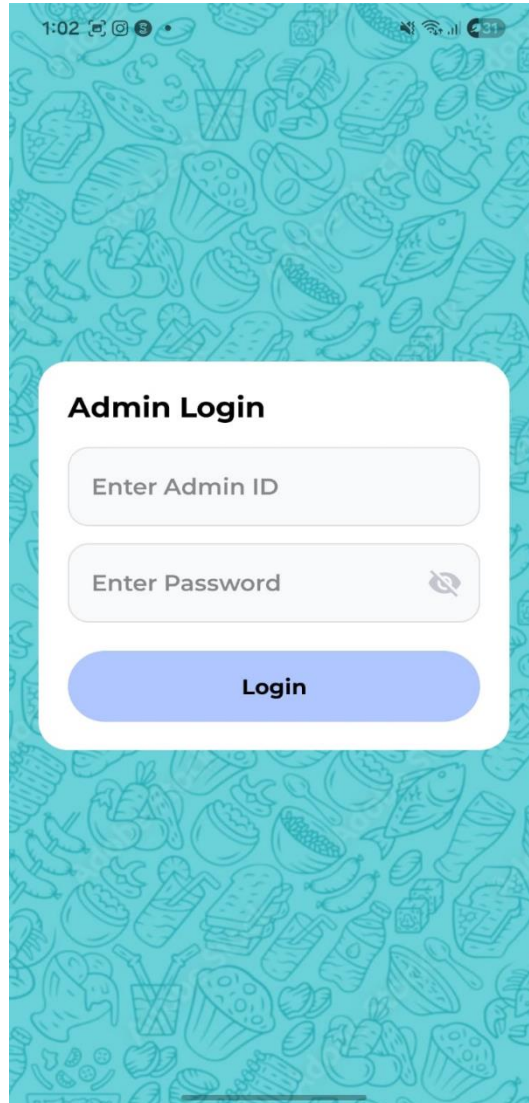
Save Changes

5.22 Admin Login Screen

The Admin Login Screen allows administrators to access advanced management features.

It includes fields for Admin ID and password, validated against predefined credentials.

Successful login redirects to the Admin Dashboard.

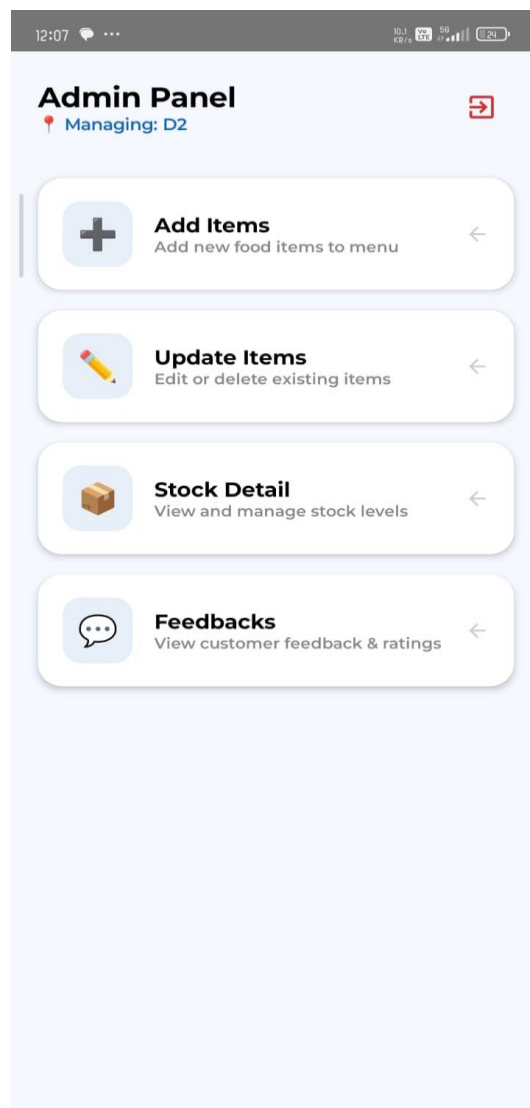
A mobile application screenshot of the Admin Login screen. The background is a teal color with a repeating pattern of white line-art food icons like burgers, fries, and drinks. In the center is a white rounded rectangle containing the login form. At the top of this rectangle is the title "Admin Login" in bold black text. Below the title are two input fields: "Enter Admin ID" and "Enter Password". The password field has a small eye icon to its right. At the bottom of the white rectangle is a blue rounded button with the text "Login" in white. The top of the screen shows a status bar with the time "1:02", signal strength, and battery level.

5.23 Admin Dashboard Screen — D2 (Daffodils 2)

The Admin Dashboard for D2 provides access to management functions for the D2 canteen.

It includes options such as Add Items, Update Items, Stock Management, and Feedback Review.

Each option navigates to a separate screen for managing D2 menu and operations.

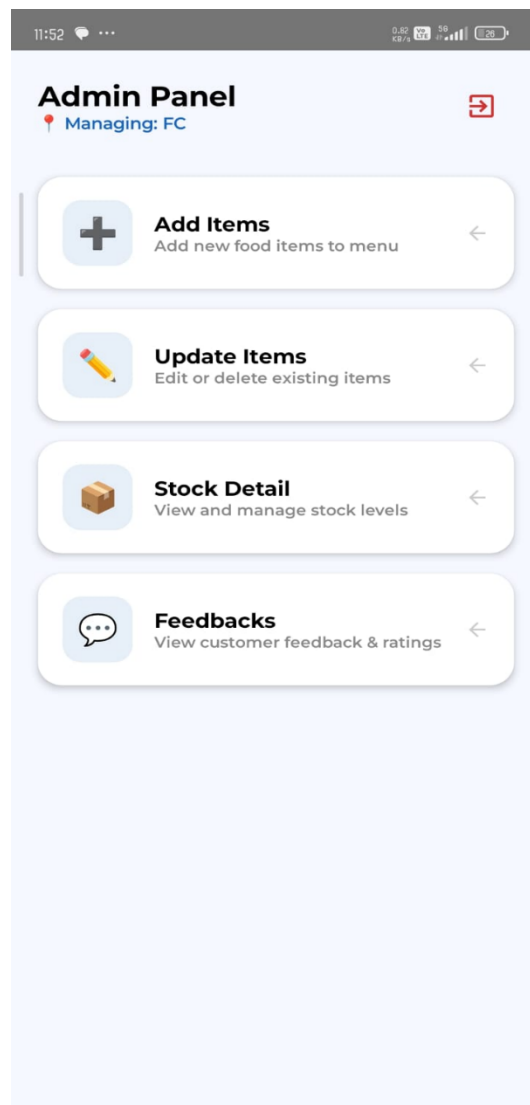


5.24 Admin Dashboard Screen — FC (Food Court)

The Admin Dashboard for FC provides access to management functions for the Food Court.

It includes options such as Add Items, Update Items, Stock Management, and Feedback Review.

Each option navigates to dedicated screens for managing FC menu items and operations.



5.25 Add Item Screen

This screen enables administrators to add new food items to the menu.

It includes fields for item details, category selection, availability, and image upload.

Data is stored in Firestore, and images are uploaded to Firebase Storage.

11:52

0.00 KB/s 58%

←

Add Item

Tap to upload image
Optional

Item Name

e.g. Chicken Biryani

Price (₹)

e.g. 120

Cooking Time (mins)

e.g. 10

Category

Veg NonVeg Egg

Tiffins Meals Biryani

Chinese Snacks Rolls

Drinks Special

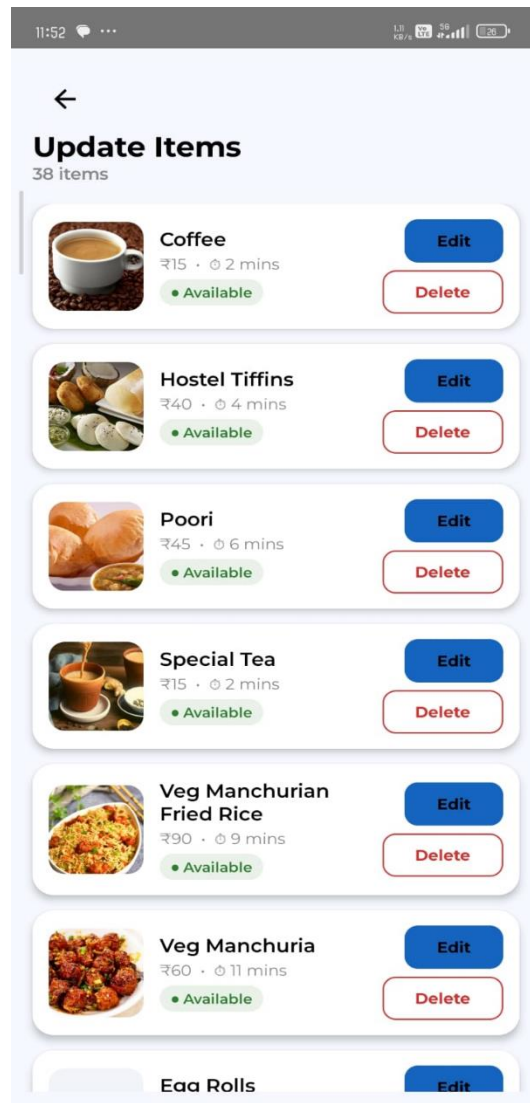
Save Item

5.26 Update Items Screen

The Update Items Screen lists all existing menu items.

Admins can edit or delete items.

Deletion requires confirmation, and changes are reflected in real time.

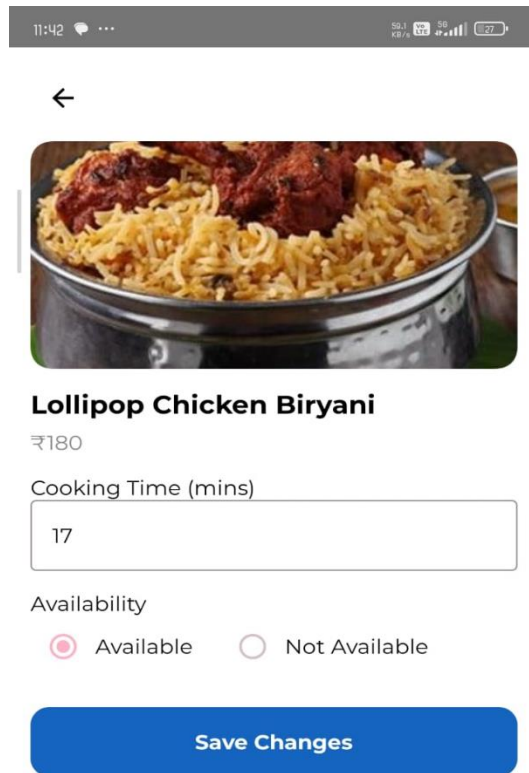


5.27 Admin Item Edit Screen

This screen allows editing of existing menu items.

Admins can modify item details and optionally update images.

Changes are saved after confirmation.



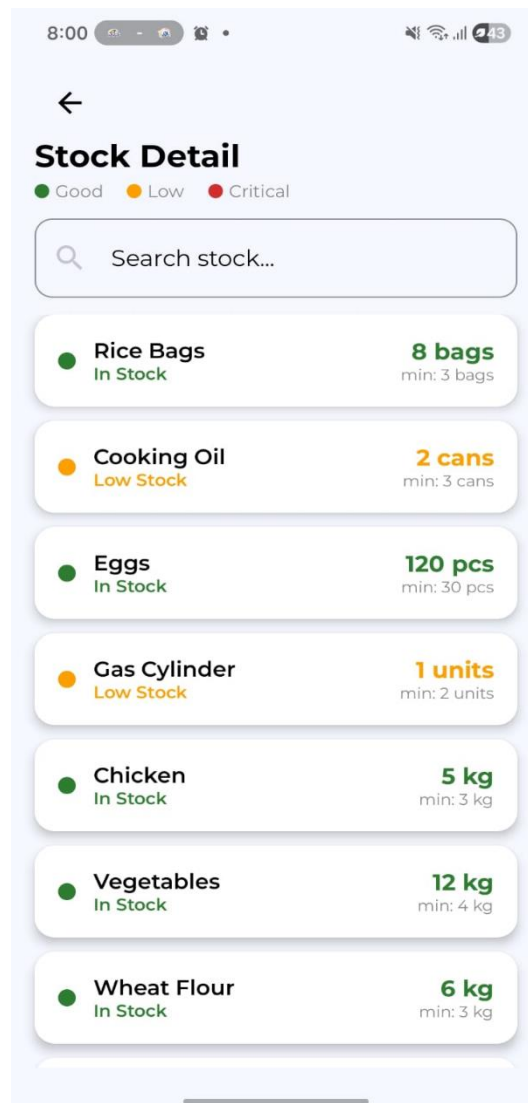
The screenshot shows a mobile application interface for editing a menu item. At the top, there is a status bar with the time 11:42, a chat icon, and network/battery indicators. Below the status bar is a back arrow icon. The main content area features a large image of a bowl of Lollipop Chicken Biryani. Below the image, the item name 'Lollipop Chicken Biryani' is displayed in bold, followed by the price '₹180'. A label 'Cooking Time (mins)' is positioned above a text input field containing the value '17'. Below the input field, the 'Availability' section shows two radio buttons: 'Available' (which is selected) and 'Not Available'. At the bottom of the screen is a blue button labeled 'Save Changes'.

5.28 Stock Detail Screen

The Stock Detail Screen provides an overview of ingredient availability.

Items are categorized as In Stock, Low Stock, or Critical using color indicators.

Stock values are currently maintained in memory and reset when the app closes.

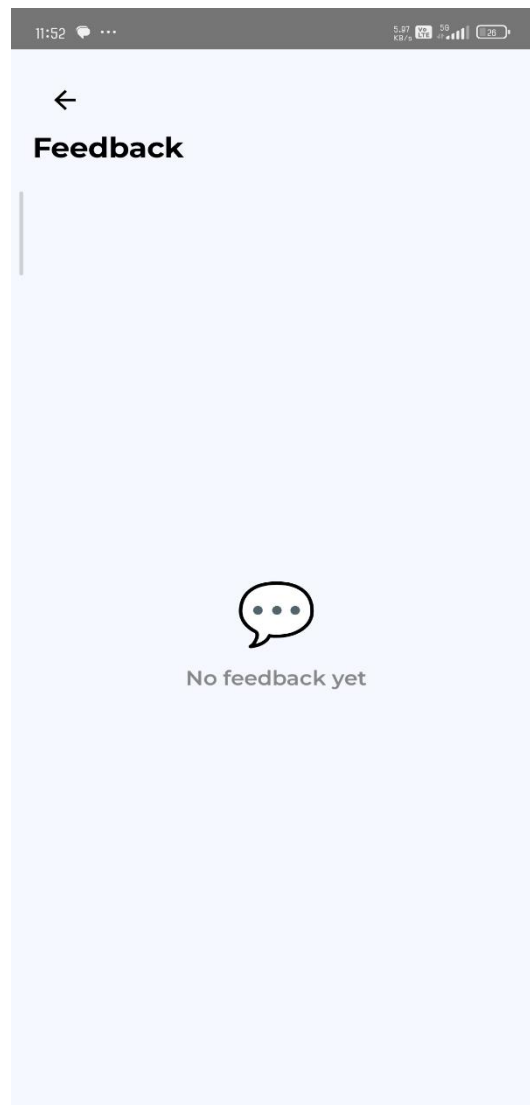


5.29 Feedback Screen (Admin)

This screen displays all customer feedback collected from users.

It shows an average rating along with individual reviews, including comments and ratings.

Feedback is fetched from Fire store and sorted by recency. An empty state is displayed when no feedback is available.



CHAPTER 6

SAMPLE CODE

CHAPTER 6

SAMPLE CODE

This chapter presents representative code samples from the Canteen Pre-Ordering System Android application developed in Kotlin. These samples illustrate the key implementation patterns used across different modules of the application.

6.1 Navigation Architecture (Core App Flow)

This shows your **state-based navigation using Jetpack Compose**

```
var currentScreen by rememberSaveable { mutableStateOf(Screen.SPLASH) }
```

```
AnimatedContent(
    targetState = currentScreen,
    transitionSpec = {
        slideInHorizontally(animationSpec = tween(400)) { it } +
            fadeIn(animationSpec = tween(400)) togetherWith
            slideOutHorizontally(animationSpec = tween(400)) { -it } +
            fadeOut(animationSpec = tween(400))
    }
) { screen ->

    when (screen) {

        Screen.SPLASH -> SplashScreen(
            onFinish = { currentScreen = Screen.LOGIN_PHONE }
        )
    }
}
```

```

Screen.LOGIN_PHONE -> PhoneLoginScreen(
    onSendOtp = { /* Firebase OTP logic */ }
)

Screen.HOME -> HomeScreen(
    onItemClick = { currentScreen = Screen.ITEM }
)

Screen.CART -> CartScreen(
    onOrderSuccess = { currentScreen = Screen.ORDER_SUCCESS }
)
}
}

```

6.2 Data Models (Kotlin Data Classes)

These represent your **core system entities used across the app**

```

data class FoodItem(
    val name: String,
    val price: Int,
    val categories: List<String>,
    val cookingTime: Int,
    val rating: Double,
    val id: String = "",
    val isAvailable: Boolean = true,
    val location: String = "D2",
    val imageUrl: String = ""
)

```

```
)
data class CartItem(
    val food: FoodItem,
    val quantity: Int,
    val note: String,
    val cookingTime: Int
)
```

```
data class Admin(
    val id: String,
    val password: String,
    val location: String
)
```

6.3 Firebase Phone OTP Authentication

Implements **Firebase Phone Authentication using OTP**

```
val options = PhoneAuthOptions.newBuilder(FirebaseAuth.getInstance())
    .setPhoneNumber(phoneNumber)
    .setTimeout(60L, TimeUnit.SECONDS)
    .setActivity(this@MainActivity)
    .setCallbacks(object : PhoneAuthProvider.OnVerificationStateChangedCallbacks() {

        override fun onVerificationCompleted(credential: PhoneAuthCredential) {
            FirebaseAuth.getInstance().signInWithCredential(credential)
        }

        override fun onVerificationFailed(e: FirebaseException) {
```

```

        Toast.makeText(
            this@MainActivity,
            "OTP Failed: ${e.message}",
            Toast.LENGTH_LONG
        ).show()
    }

    override fun onCodeSent(
        verificationIdParam: String,
        token: PhoneAuthProvider.ForceResendingToken
    ) {
        verificationId = verificationIdParam
        currentScreen = Screen.LOGIN_OTP
    }
})
.build()

```

```
PhoneAuthProvider.verifyPhoneNumber(options)
```

6.4 Firestore Real-Time Menu Loading

Provides **real-time data synchronization from Firebase Firestore**

```
LaunchedEffect(Unit) {
```

```

val db = FirebaseFirestore.getInstance()

db.collection("items")
    .addSnapshotListener { snapshot, error ->

        if (error != null) return@addSnapshotListener

        val list = snapshot?.documents?.mapNotNull { doc ->

            val name = doc.getString("name") ?: return@mapNotNull null

            FoodItem(
                id = doc.id,
                name = name,
                price = (doc.getLong("price") ?: 0).toInt(),
                categories = doc.get("category") as? List<String> ?: listOf("Other"),
                cookingTime = (doc.getLong("cookingTime") ?: 5).toInt(),
                rating = 4.0,
                isAvailable = doc.getBoolean("isAvailable") ?: true,
                location = doc.getString("location") ?: "D2",
                imageUrl = doc.getString("imageUrl") ?: ""
            )

        } ?: emptyList()

        firebaseItems = list

```

```
    }  
}
```

6.5 Add to Cart Logic

Handles cart item merging and quantity updates

```
fun addToCart(  
    food: FoodItem,  
    qty: Int,  
    note: String,  
    cartItems: List<CartItem>  
) : List<CartItem> {  
  
    val cleanNote = note.trim()  
  
    val existing = cartItems.find {  
        it.food.name == food.name && it.note == cleanNote  
    }  
  
    return if (existing != null) {  
        cartItems.map {  
            if (it.food.name == food.name && it.note == cleanNote) {  
                it.copy(quantity = it.quantity + qty)  
            } else it  
        }  
    } else {  
        cartItems + CartItem(  
            food = food,  

```

```

        quantity = qty,
        note = cleanNote,
        cookingTime = food.cookingTime
    )
}
}

```

6.6 Order ID Generation

Generates **unique order IDs** for each transaction

```

fun generateOrderId(location: String, total: Int): String {
    val time = System.currentTimeMillis().toString().takeLast(6)
    val locCode = if (location == "D2") "D2" else "FC"
    return "BVRN-$locCode-$time-$total"
}

```

6.7 UPI Payment Integration

Implements UPI-based payment using Android Intent

```

fun payWithUpi(
    context: android.content.Context,
    amount: Int,
    upiId: String,
    packageName: String? = null
) {

```

```

        val uri = "upi://pay?pa=$upiId&pn=BVRIT Smart  
Canteen&am=$amount&cu=INR".toUri()

```

```

        val intent = Intent(Intent.ACTION_VIEW, uri)

```

```

        if (packageName != null) {
            intent.setPackage(packageName)
        }

```

```

        context.startActivity(intent)
    }

```

6.8 QR Code Generation

Used for order verification at the canteen counter

```

fun generateQrCode(data: String): ImageBitmap {

    val writer = QRCodeWriter()

    val bitMatrix = writer.encode(data, BarcodeFormat.QR_CODE, 512, 512)

    val bitmap = createBitmap(512, 512, Bitmap.Config.RGB_565)

    for (x in 0 until 512) {
        for (y in 0 until 512) {
            bitmap[x, y] = if (bitMatrix[x, y])
                android.graphics.Color.BLACK

```

```

        else android.graphics.Color.WHITE
    }
}

return bitmap.asImageBitmap()
}

```

6.9 Real-Time Order Tracking Listener

Enables **live order status updates using Firestore listeners**

```

LaunchedEffect(activeOrderId) {
    if (activeOrderId.isNotEmpty()) {
        val db = FirebaseFirestore.getInstance()
        db.collection("orders")
            .document(activeOrderId)
            .addSnapshotListener { snapshot, _ ->
                if (snapshot != null && snapshot.exists()) {
                    val data = snapshot.data?.toMutableMap()
                    if (data != null) {
                        data["docId"] = snapshot.id
                        activeOrderData = data
                    }
                }
            }
    }
}

```

CHAPTER 7

CONCLUSION

CHAPTER 7

CONCLUSION

The Canteen Pre-Ordering System for Campus Food Courts has been successfully designed, developed, and partially deployed as a functional Android application for B.V. Raju Institute of Technology. The application addresses the fundamental problem of long queues and operational inefficiency at the campus canteens — Daffodils 2 (D2) and Food Court (FC) — by providing a digital pre-ordering platform that is accessible, intuitive, and efficient.

Through the course of this project, the development team has successfully implemented the core functionalities of the system including user authentication via Firebase, real-time menu browsing for both D2 and FC canteens, cart management, order placement with unique order ID generation, real-time order tracking, order history, and an admin panel for canteen staff to manage incoming orders and update order statuses.

The application is built on a solid technical foundation using Kotlin programming language, Android Studio IDE, MVVM architecture pattern, and Google Firebase as the backend-as-a-service platform. This combination of technologies ensures that the application is maintainable, scalable, and aligned with current industry best practices in Android development.

The project has demonstrated that a campus-specific food ordering system can be effectively built using modern mobile development frameworks without requiring a complex server-side infrastructure. Firebase's real-time database capabilities, combined with Kotlin's expressive syntax and MVVM's clean architecture, have proven to be an excellent technology stack for this use case.

From a functional perspective, the system successfully reduces the ordering time for students from an average of 20-30 minutes (including queue time) to less than 5 minutes for order placement, with food being ready for pickup when the student arrives at the canteen. This represents a significant improvement in the campus dining experience and aligns with the project's primary objective.

The project has also been a valuable learning experience for the team. The practical application of concepts such as Firebase integration, MVVM architecture, RecyclerView adapters, LiveData and ViewModel lifecycle management, Kotlin coroutines, and Android Navigation Component has significantly enhanced the team's proficiency in Android development. The project also reinforced important software engineering principles such as modular design, separation of concerns, iterative development, and version control using Git.

In conclusion, the Canteen Pre-Ordering System represents a meaningful contribution to the digitization of campus facilities at BVRIT. While the current version has certain limitations, the core system is functional, tested, and ready for full-scale deployment. The groundwork laid in this project provides a strong foundation for future enhancements that can further improve the system's capabilities and user experience.

CHAPTER 8

FUTURE WORK

CHAPTER 8

FUTURE WORK

While the current version of the Canteen Pre-Ordering System delivers the core pre-ordering functionality, several enhancements and new features have been identified for future development cycles. These improvements would significantly extend the system's capabilities, improve the user experience, and broaden the application's impact on campus food service operations.

8.1 iOS Application Development

Developing a companion iOS application using Swift or a cross-platform version using Flutter would extend the system's reach to Apple device users within the campus community. A Flutter-based cross-platform solution would allow maintaining a single codebase for both Android and iOS, significantly reducing development and maintenance overhead.

8.2 Push Notifications

Implementing push notifications using Firebase Cloud Messaging (FCM) would greatly improve the order tracking experience. Instead of requiring users to manually check the app for status updates, the system would automatically send push notifications when an order's status changes (e.g., 'Your order #1234 is being prepared' and 'Your order #1234 is ready for pickup!'). This would allow students to resume their academic activities and be notified only when their food is ready.

8.3 Institutional Email Verification

Restricting registration to verified BVRIT institutional email addresses (ending with @bvrit.ac.in) would ensure that only legitimate campus members can use the application. This can be implemented by validating the email domain during registration and potentially integrating with the college's student information system for automatic student verification.

8.4 Advanced Analytics Dashboard

Developing an analytics dashboard for canteen management would provide valuable insights into ordering patterns, peak hours, popular items, and revenue trends. This data could be visualized through charts and graphs, enabling evidence-based decision making for menu planning, staff scheduling, and inventory management. The analytics module could be built using MP Android Chart library for in-app visualization or integrated with Google Looker Studio for web-based reporting.

8.5 Inventory Management System

Integrating a real-time inventory management module would allow canteen staff to track ingredient and item quantities. As orders are placed, the system would automatically deduct consumed items from the inventory, trigger low-stock alerts, and automatically mark items as unavailable when stock runs out. This would prevent situations where students order items that are not physically available.

8.6 Scheduled Pre-Ordering

Allowing students to place orders in advance for a specific pickup time (e.g., ordering breakfast in the evening for the next morning, or ordering lunch during a morning class) would maximize the operational benefits of the pre-ordering system. Scheduled orders would give canteen staff even more preparation time and allow for better resource planning.

8.7 Ratings and Reviews

Adding a post-delivery ratings and review system would allow students to rate their dining experience and provide feedback on specific menu items. This data would help canteen management identify popular items, improve underperforming dishes, and maintain quality standards. The rating data could also be displayed on item cards to help other students make informed choices.

8.8 Multi-Campus Expansion

The system architecture has been designed with scalability in mind, and the platform can be expanded to support additional campus canteens or even multiple college campuses under the same organization. The database structure uses canteen IDs as keys, making it straightforward to add new canteens without significant architectural changes. Future versions could serve as a unified campus food ordering platform across multiple institutions.

8.9 AI-Powered Menu Recommendations

Analysing each student's order history using on-device machine learning (TensorFlow Lite) to generate personalised food recommendations — "You usually order Dosa on weekday mornings" — would increase order value and improve student engagement with the digital menu

8.10 Offline Mode with Background Sync

Implementing an offline-first architecture using Android Room for local caching, with WorkManager managing background synchronisation when internet connectivity is restored, would allow students to browse the menu and build a cart even under poor network conditions common in large campus buildings.

8.11. Demand Forecasting and Analytics Dashboard

Implementing demand forecasting algorithms on historical Firestore order data would enable canteen administrators to predict which items will be popular on a given day, optimise

ingredient procurement, and reduce food wastage — addressing one of the root inefficiencies identified in the problem statement.

CHAPTER 9

REFERENCES AND CITATIONS

CHAPTER 9

REFERENCES AND CITATIONS

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